



IEEE Standard for Synchrophasor Data Transfer for Power Systems

IEEE Power and Energy Society

Developed by the
Power System Communications and Cybersecurity
Committee

IEEE Std C37.118.2™-2024
(Revision of IEEE Std C37.118.2-2011)



IEEE Standard for Synchrophasor Data Transfer for Power Systems

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Power System Communications and Cybersecurity Committee
of the
IEEE Power and Energy Society

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IEEE SA Standards Board

Abstract: A method for the real-time exchange of synchronized phasor measurement data between power system equipment is defined within this standard. This includes the following: messaging that can be used with any suitable communication protocol for real-time communication between phasor measurement units (PMU), phasor data concentrators (PDC), and other applications; message types, contents, and their use as well as data types and format specifications; descriptions of typical measurement systems; communication options and requirements. Note that while cybersecurity is discussed, and while its importance is acknowledged in the overall system, it is considered beyond the scope of this standard.

Keywords: IEEE C37.118.2™, PDC, phasor, phasor data concentrator, phasor measurement unit, PMU, synchronized phasor, synchrophasor

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Introduction

This introduction is not part of IEEE Std C37.118.2-2024, IEEE Standard for Synchrophasor Data Transfer for Power Systems.

Digital computer-based measurement, protection, and control systems have become common features of electric power systems. These measurement, protection, and control systems use sampled data to compute various quantities such as voltage and current phasors. Phasors are used in many protection and data acquisition functions, and their utility is increased further by referencing them to a common time base. This can be accomplished by synchronizing the phasor estimate to a precise time source that is common to the various measuring sites. Phasor estimates synchronized to a common time source and referenced to a common frequency are defined as synchrophasors. Simultaneous measurement sets derived from synchronized phasors provide a vastly improved method for tracking power system dynamic phenomena for improved power system monitoring, protection, operation, and control.

The present version of the standard is the fourth revision of this standard. The original standard for synchrophasors was IEEE Std 1344™-1995 [B4], which was reaffirmed in 2001.⁶ This standard was replaced in 2005 by IEEE Std C37.118™-2005 [B5], and in 2011 by IEEE Std C37.118.1™-2011 [B6] and IEEE Std C37.118.2™-2011 [B7]. IEEE Std C37.118-2005 provided additional clarification for the phasor and synchronized phasor definitions. The concepts of total vector error and compliance tests were introduced. The message formats were updated from the original standard to improve information exchange with other systems such as a master station. Specifically, the sync, frame size, and station identification fields were added to the data frame, configuration frame, header frame, and command frame. Analog data was added to the data frame, the fraction-of-second (FRACSEC) field replaced the sample count field, and the status field was significantly modified to include time quality.

The 2005 version of IEEE Std C37.118 included both measurement requirements and real-time data transfer requirements. To simplify widespread adoption and facilitate the use of other communication protocols for phasor data transfer, IEEE Std C37.118-2005 [B5] was split into two standards: IEEE Std C37.118.1™-2011 [B6] for synchrophasor measurement requirements, and IEEE Std C37.118.2-2011 [B7] for synchrophasor data transfer requirements. The 2011 edition of C37.118.1 added definitions, tests, and error limits for measurement of dynamically changing signals. It also added evaluation of frequency and ROCOF measurement under all the test conditions. C37.118.2 retained the previous requirements for data transfer, and one new configuration frame was added.

Wide usage over several years revealed several shortcomings and ambiguities in IEEE Std C37.118.2-2011 [B7]. This revision corrects these. This will improve interoperability between devices and applications from different vendors. Significant changes in this revision include discrete event data frames, measurement-based data attributes flags, separate time quality flags, and remote configuration. Frequency and rate-of-change-of-frequency are not mandatory anymore.

Numerous devices and systems compliant with the 2011 version of the standard have been in use for several years, and will continue to be for many more. This version of the standard includes support for the 2011 version, so existing systems can remain compliant. There will be a need for the two versions to co-exist, and for systems to be able to convert and combine between the two. This release includes some guidance to help accomplish this.

⁶The numbers in brackets correspond to those of the bibliography in Annex K.

Contents

1. Overview	13
1.1 Scope	13
1.2 Purpose	13
1.3 Word usage	13
2. Normative references	14
3. Definitions, acronyms, and abbreviations	14
3.1 Definitions	14
3.2 Acronyms and abbreviations	15
4. Synchrophasor messaging	16
4.1 Messaging overview	16
4.2 Types of messages	16
4.3 Message framework	17
4.4 Configuration frames	20
4.5 Data frames	25
4.6 Commands	30
4.7 Error-response frame	34
5. Communications	36
5.1 Periodic data transmission	37
5.2 Discrete event frames transmission	37
Annex A (normative) Version 2 compliance requirements (optional)	38
Annex B (informative) Synchrophasor measurements	50
Annex C (informative) Synchrophasor measurement systems	53
Annex D (informative) System communications considerations	56
Annex E (informative) Configuration example	60
Annex F (informative) Leap second bit timing examples	62
Annex G (informative) FRACSEC/TIME_BASE usage examples	63
Annex H (informative) Usage of mixed version PMUs	65
Annex I (informative) CONT_IDX handling example	73
Annex J (informative) XML format config frames	75
Annex K (informative) Bibliography	77

List of Figures

Figure 1—An example of a synchrophasor system	16
Figure 2—Example of frame transmission order.....	18
Figure A.1—Example of frame transmission order for version 2	38
Figure B.1—Convention for synchrophasor representation.....	51
Figure C.1—Synchrophasor data collection network.....	53
Figure D.1—Mapping of IEEE C37.118 data into a TCP or UDP packet—A transport layer header and trailer are shown, as it would be when using Ethernet.....	58
Figure E.1—Configuration command structure	60
Figure H.1—Substation PDC using version 2, control center PDC version irrelevant	66
Figure H.2—Control Center PDC using version 2	66
Figure H.3—Control center PDC using version 3	69

List of Tables

Table 1—Word definitions common to all frame types.....	18
Table 2—Name field description.....	21
Table 3—Configuration frame format.....	21
Table 4—Periodic data frame format.....	26
Table 5—STAT_FLAG WORD.....	27
Table 6—TIMEQUALITY field format for version 3 data stream.....	28
Table 7—TIMEQUALITY field format for a version 2 data stream conversion to a version 3 data stream....	28
Table 8—Data attributes bit definitions.....	29
Table 9—Discrete events data frame format—all digitals.....	29
Table 10—Command frame format.....	30
Table 11—Commands sent to the PMU/PDC.....	30
Table 12—Old data request command.....	31
Table 13—Stream ID available command.....	32
Table 14—Rename signals frame format.....	32
Table 15—Configure stream frame format.....	33
Table 16—Error-response frame description.....	34
Table 17—Error-response words 1 and 2.....	35
Table 18—Required PMU reporting rates.....	37
Table A.1—Word definitions common to all frame types for version 2.....	38
Table A.2—Time quality flag bit definitions for version 2.....	39
Table A.3—4-bit message time quality indication codes (MSG_TQ) for version 2.....	39
Table A.4—Data frame organization for version 2.....	40
Table A.5—Word definitions unique to data frames for version 2.....	40
Table A.6—3-bit PMU time quality indication codes (PMU_TQ) for version 2.....	42
Table A.7—Configuration frame 1 and 2 organization, for version 2.....	42
Table A.8—Word definitions unique to configuration frames 1 and 2, for version 2.....	43
Table A.9—Configuration frame 3 organization, for version 2.....	44
Table A.10—Word definitions unique to configuration frame 3.....	45
Table A.11—Name field description.....	48

Table A.12—Header frame organization, for version 2	48
Table A.13—Command frame organization, for version 2	49
Table A.14—Commands sent to the PMU/PDC, for version 2	49
Table D.1—Estimation of bandwidth requirements for example PMU configurations	56
Table D.2—Summary of delays' causes and typical ranges.....	57
Table E.1—A simple step by step configuration method	60
Table G.1—Rounding-off of FRACSEC values (30 FPS / 60 Hz)	63
Table G.2—Rounding-off of FRACSEC values (25 FPS / 50 Hz)	63
Table G.3—Discrete event FRACSEC examples for 50 Hz system	64
Table G.4—Discrete event FRACSEC examples for 60 Hz system	64
Table H.1—Commands conversion from version 2 to version 3	66
Table H.2—Configuration frame conversion from version 3 to version 2	67
Table H.3—Data frame conversion from version 3 to version 2	68
Table H.4—Partial conversion of version 3 STAT_FLAG word to version 2	68
Table H.5—Commands conversion from version 3 to version 2	69
Table H.6—Configuration frame conversion from version 2 to version 3	69
Table H.7—Configuration frame conversion from version 2 to version 3	70
Table H.8—Data frame conversion from version 2 to version 3	72
Table H.9—Partial conversion of version 2 STAT word to version 3	72

IEEE Standard for Synchrophasor Data Transfer for Power Systems

1. Overview

1.1 Scope

This standard defines a method for the exchange of synchronized phasor measurement data between power system equipment. It specifies messaging including types, use, contents, and data formats for real-time communication between phasor measurement units (PMU), phasor data concentrators (PDC), and other applications. This revision includes the following:

- a) Communications between a single PMU and a PDC,
- b) PDC functions dealing with multiple PMUs, and
- c) Onwards communications from the PDC.

1.2 Purpose

The purpose for this standard is to facilitate data exchange among measurement, data collection, and application equipment. It provides a defined, open access method for all vendors to use to facilitate development and use of synchrophasors. It is a simple and direct method for data transmission within a phasor measurement system which may be used directly or with other communication protocols. This method was initially established by IEEE Std C37.118-2005 [B5], Standard for Synchrophasors for Power Systems, and subsequently by IEEE Std C37.118.2-2011 [B7].

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall equals is required to*).^{7,8}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should equals is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may equals is permitted to*).

⁷The use of the word *must* is deprecated and cannot be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.

⁸The use of *will* is deprecated and cannot be used when stating mandatory requirements; *will* is only used in statements of fact.

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

The word “reserved” indicates:

- If used in an assignment of values to an enumeration or an attribute that the values indicated are reserved for use in future editions of this standard and shall not be used for any other purpose.
- If used in a field of a message or frame that the field is reserved for use in future editions of this standard. The field shall be present in the message or frame, with the size specified. No interpretation of reserved fields is to be made for this edition of this standard, and the fields shall not be used for any other purpose.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEC/IEEE Std 60255-118-1™, IEC/IEEE Standard, Measuring relays and protection equipment—part 118-1: Synchrophasor for power system – Measurements.^{9,10}

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this Clause.¹¹

command frame: A machine-readable message sent to a phasor measurement unit (PMU) or a phasor data concentrator (PDC) for control or configuration.

configuration frame: A machine-readable message describing the data types, calibration factors, and other meta-data for the data that a PMU or a PDC sends.

coordinated universal time (UTC): An internationally agreed upon time scale maintained by the Bureau International des Poids et Mesures (BIPM) used as the time reference by nearly all commonly available time and frequency distribution systems (e.g., WWV, WWVH, LORAN-C, Transit, Omega, and GPS).

CRC-CCITT: 16-bit cyclic redundancy check (CRC) calculated using the generating polynomial $X^{16} + X^{12} + X^5 + 1$, seed value 0xFFFF (−1), no final mask.

data frame: A set of synchrophasor, frequency, rate-of-change-of-frequency, analog, and digital measurements that corresponds to the same time stamp.

NOTE—The term “frame” is used to differentiate it from individual measured values.¹²

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¹¹The *IEEE Standards Dictionary: Glossary of Terms & Definitions* is available at: <http://dictionary.ieee.org>. An IEEE account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

¹²Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

leap second: A positive or negative one-second adjustment to the coordinated universal time that keeps it close to mean solar time.

phasor: Complex equivalent of a sinusoidal wave quantity such that the complex modulus is the cosine wave amplitude, and the complex angle (in polar form) is the cosine wave phase angle.¹³

phasor data concentrator (PDC): A function that collects phasor data, and discrete event data from phasor measurement units and possibly from other PDCs and transmits data to other applications.

phasor measurement unit (PMU): Device or function in a multifunction device that produces synchronized phasor, frequency, and rate-of-change-of-frequency (ROCOF) estimates from voltage and/or current signals and a time synchronizing signal.¹⁴

synchronized phasor: *See: synchrophasor.*

synchrophasor (synchronized phasor): Phasor representing the fundamental of an AC signal whose magnitude is the rms value of the fundamental amplitude and angle is the difference between the signal fundamental angle and the phase angle of a cosine at the nominal signal frequency that is synchronized to coordinated universal (UTC) time.¹⁵

NOTE—In this standard, the phasors from remote sites have a defined common phase relationship.

wait time: The maximum time interval during which the phasor data concentrator collects data for a particular output from one or more inputs. It is the time interval added to a message time stamp time or message-arrival time after which no more data with that time stamp is accepted.¹⁶

3.2 Acronyms and abbreviations

CRC	cyclic redundancy check
DADFREQ	data attributes associated with rate-of-change-of-frequency measurement
DAFREQ	data attributes associated with frequency measurement
DAPHASOR	data attributes associated with phasor measurement
DFR	digital fault recorder
FRACSEC	fraction-of-second
IP	internet protocol (see IETF RFC 791 [B13])
ROCOF	rate-of-change-of-frequency
SOC	second of century
TCP	transmission control protocol (see IETF RFC 793 [B14])
UDP	user datagram protocol (see IETF RFC 768 [B12])
UTF-8	unicode character encoding scheme used to represent text in writing systems

¹³Repeated from IEC/IEEE Std 60255-118-1™.

¹⁴Repeated from IEC/IEEE Std 60255-118-1™.

¹⁵Repeated from IEC/IEEE Std 60255-118-1™.

¹⁶Repeated from IEEE Std C37.247™-2019 [B11].

4. Synchrophasor messaging

This Clause describes the format of messages to and from a PMU or PDC for use in real-time communication of phasor data. *Real-time data transmission* here is defined as taking place concurrently with the measurement process and will include measurement delays and transmission delays. This standard does not address requirements for stored phasor data. Implementation of additional protocols for phasor data communication is not restricted.

4.1 Messaging overview

Messages and formats specified in this Clause are referred to as version 3 of the standard (See [Table 1](#)). Messages and formats compliant with the 2011 version (version 2) of this standard are included in [Annex A](#). All requirements for devices and applications shall be specified by the version number. A device/application shall meet all the requirements as specified for version 2 or version 3 to be considered as compliant with this standard. Compliance specification shall include the version number. [Annex H](#) includes guidance for use of both versions in the same synchrophasor measurement system.

Any communication system or media capable of carrying the messages defined in this standard may be used for transmission of these messages. For compliance, these messages shall be transmitted in their entirety as they are specified.

A general description of synchronized measurements, measurement systems, and communications considerations is included in [Annex B](#), [Annex C](#), and [Annex D](#). An example of a simple synchrophasor measurement system is shown in [Figure 1](#).

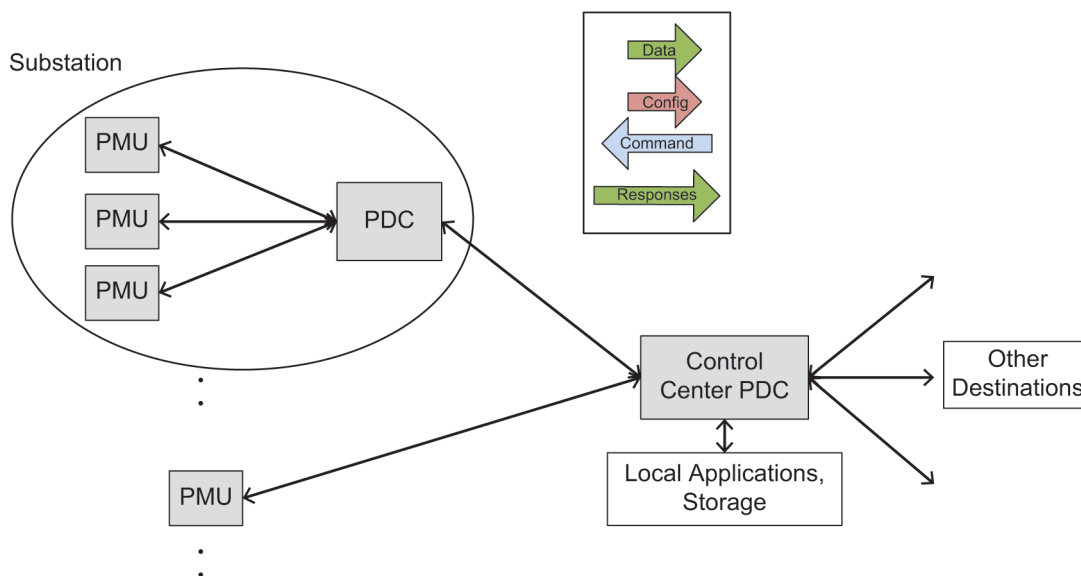


Figure 1—An example of a synchrophasor system

4.2 Types of messages

Five message types are defined: *configuration*, *periodic data*, *discrete event data*, *command*, and *error-response*.¹⁷ Each message type uses a specific machine-readable frame format. Since all messages are sent as frames, the words “message” and “frame” are used interchangeably in this standard as follows:

¹⁷The *header* message defined in previous versions of this standard is deprecated.

- Configuration messages include the data types, calibration factors, and other meta-data for the data that the PMU/PDC sends. Two configuration messages are defined in this standard: capability and stream configuration.
- Data messages include the measurements made by a PMU. Two types of data messages are defined: periodic data messages and discrete event data messages.
- Commands are messages sent to the PMU/PDC (that is the data source) for control or configuration. Two new optional configuration command messages are defined in this standard: configure stream and rename signals. A device or an application does not have to implement these commands to be considered compliant with this standard.
- Error-response messages are sent by the data source PMU/PDC in reply to commands when a command is not understood or rejected (such as for an unimplemented feature).

Periodic data messages are sent at regular time intervals determined by the data reporting rate (see 5.1.1). Discrete event data messages may be sent at any time, usually to report a discrete event such as a change in a digital signal. Signals included in discrete event data messages may also be included in periodic data messages. Data messages do not include any metadata that describe the data content. Such metadata are included in configuration messages. The match between configuration and data is achieved by a `STREAM_ID` (see 4.3.1.3). The same `STREAM_ID` is also used by command and *error*-response messages. All such messages between a data source (a PMU or a PDC) and a data receiver (usually another PDC), using the `STREAM_ID`, are identified as a data stream.

A PMU or PDC may transmit multiple data streams, each with different content, rate, format, etc. Each data stream shall have its own `STREAM_ID` so that the data, configuration, command, and *error*-response messages can be identified.

Information may be stored in any convenient form in the PMU/PDC itself, but when transmitted as messages, it shall be formatted as frames described from 4.3 to 4.7. Commands and other messages received by a connected device that are not understood may generate an *error*-response (see 4.7). This revision includes capability for retrieval of historical data.

In normal operation, the PMU sends data frames continuously. Configuration and *error*-response frames are sent on receiving the corresponding command. Configuration frames may also be sent spontaneously. This is useful in systems that do not permit commands.

4.3 Message framework

4.3.1 Overall message

All message frames consist of the following fields in sequence:

- 2-B SYNC word.
- 2-B FRAMESIZE word.
- 2-B `STREAM_ID`.
- 8-B time stamp consisting of a 4-B second-of-century (SOC), 1-B leap second information (LEAP_BYTE), and 3-B fraction of a second (FRACSEC).
- Multiple data fields. The number and format of such fields depends on the type of message sent.
- 2-B check word (CHK). This is a CRC-CCITT, 16-bit unsigned integer. It is calculated over all bytes in the frame, from SYNC, until the last data bytes inclusive.

All frames are transmitted exactly as described with no delimiters. Figure 2 illustrates the frame transmission order. The SYNC word is transmitted first and CHK word last. Two- and four-byte words including integer and floating-point numbers are transmitted most significant byte first (network or “big endian” order). All frame types use this same order and format.

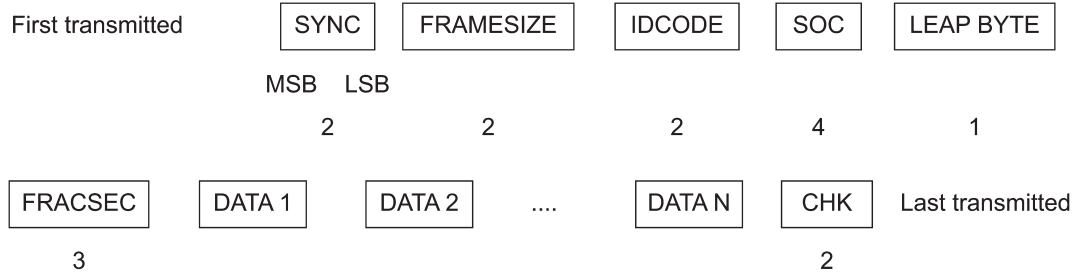


Figure 2—Example of frame transmission order

Word definitions common to all data frames are described in Table 1. Changes in this version include support for this version of the standard in the SYNC field, and separation of LEAP_BYTE field from the FRACSEC field.

Table 1—Word definitions common to all frame types

No.	Field	Size (bytes)	Comments
1	SYNC	2	Frame synchronization word. Leading byte: AA hex Second byte: Frame type and version, divided as follows:^a Bits 7 to 4: For version 3 (this version of the standard) 1000: Periodic data frame (defined in 4.5.1) 1001: Discrete event data frame (defined in 4.5.2) 1010: Capability frame (defined in 4.4.1.1) 1011: Stream configuration frame (defined in 4.4.1.2) 1100: Command frame (defined in 4.6.1) 1101: Rename signals frame (command) (defined in 4.6.2.1, optional) 1110: Configure stream frame (command) (defined in 4.6.2.2, optional) 1111: Error-response frame (defined in 4.7) Bits 3 to 0: Version number, in binary (1 to 15) Version 3 (0011) for messages defined in this revision.
2	FRAMESIZE	2	Total number of bytes in the transmitted frame, including CHK. 16-bit unsigned integer. Range = maximum 65 535.
3	STREAM_ID	2	Data stream ID number, 16-bit unsigned integer, assigned by user, 1 to 65534 (0 and 65535 are reserved). Identifies destination data stream for commands and source data stream for other messages. A stream will be hosted by a device that can be physical or virtual. If a device only hosts one data stream, the STREAM_ID identifies the device as well as the stream. If the device hosts more than one data stream, there shall be a different STREAM_ID for each stream. This may be a PMU data stream or a PDC data stream.
4	SOC	4	Time stamp, 32-bit unsigned number, SOC count starting at midnight 01-Jan-1970 (UNIX time base). Range is 136 years, rolls over 2106 AD. Leap seconds are not included in count, so each year has the same number of seconds except leap years, which have an extra day (86 400 s).

Table continues

Table 1—Word definitions common to all frame types (continued)

No.	Field	Size (bytes)	Comments
5	LEAP_BYTE	1	Leap second information. If leap seconds are eliminated from UTC, bits 4 to 6 shall be set to 0. Bit 7: Reserved. Set to 0. Bit 6: Leap second direction. 0 = add, 1 = delete. Shall be set at the same time or before the leap second pending bit is set and remains unchanged for 24 h after the leap second occurs. Bit 5: Leap second occurred. Shall be set in the first second after the leap second occurs and remains set for 24 h. Bit 4: Leap second pending. Shall be set not more than 12 h nor less than 1 s before a leap second occurs and be cleared in the second after the leap second occurs. Bits 3 to 0: Reserved. Set to 0. ^b Examples of leap second bit timing are shown in Annex F .
6	FRACSEC	3	Fraction of second time of measurement for all data frames or time of frame transmission for non-data frames. It is an unsigned 24-bit integer number. Fractional part of a second, multiplied by TIME_BASE, and rounded to the nearest integer, is equal to FRACSEC. This calculation is shown in 4.3.1.4 . Also see Annex G .
7	CHK	2	CRC-CCITT, 16-bit unsigned integer.

^aPrevious versions of this standard required bit 7 to be reserved and set to 0. This standard has set this bit to 1 to allow for backward compatibility.

^bThese bits were used as time quality indication codes in previous versions for the entire data stream. Time quality field has been added for individual PMUs in this version (see [4.5.1.2](#)).

4.3.1.1 SYNC word

The SYNC word provides synchronization and frame identification. This word is detailed in [Table 1](#). Bits 7 to 4 in the SYNC word designate the frame type. Bits 3 to 0 describe version number. Since this is the third edition of the synchrophasor standard, new messages shall be designated version 3 (binary 0011). All previously defined messages are unchanged in this standard and shall remain version 1 or 2 as applicable.

4.3.1.2 FRAMESIZE

The FRAMESIZE field shows the number of bytes in the entire frame, from the SYNC word to the CHK word, both inclusive.

4.3.1.3 STREAM_ID

The STREAM_ID identifies the source of a data, configuration, or *error*-response message, and the destination of a command message. The message STREAM_ID is associated with a data stream and ties data frames with the associated configuration information. The STREAM_ID for each output stream shall be user settable.

4.3.1.4 TIME STAMP

The time stamp consists of a 4-B second of century (SOC) field, a 1-B leap second information (LEAP_BYTE) field, and a 3-B fraction of second count (FRACSEC) field.

The FRACSEC indicates the nearest time to the actual time stamp that can be represented by an integer. For periodic data frames, it shall be interpreted as the nearest fractional second to the calculated time as determined by the reporting rate.

For calculating FRACSEC from the actual time at the PMU, and for calculating the time from the FRACSEC value received at the data destination, use the formulas shown below (also see [Annex G](#)).

$$\text{FRACSEC} = \text{ROUND}((\text{fractional second of time stamp}) \times \text{TIME_BASE})$$

$$\text{Time} = \text{SOC} + \text{FRACSEC} / \text{TIME_BASE}$$

4.3.1.5 CHK word

All frames terminate in check word (CHK). This is a CRC-CCITT. It uses the generating polynomial $X^{16} + X^{12} + X^5 + 1$ with an initial value of -1 (hex FFFF) and no final mask.

4.4 Configuration frames

A configuration frame is a data set containing information and processing parameters for a synchrophasor data stream. Configuration frame types are specified and identified by bits 7 to 4 of the SYNC word as shown in [Table 1](#). Optionally, XML formatted configuration frames may also be used. A possible format is shown in [Annex J](#).

4.4.1 Configuration frame changes in this version

Two configuration frames (capability, and stream configuration) are specified in this version of the standard (see [Table 1](#)).

4.4.1.1 Capability

This frame serves the same function as the previous CFG-1 frame. It shall include the information for all the signals available for transmission from a data source. This frame shall be transmitted from a data source (PMU or PDC) to the data destination (another PDC or an application). It is intended to provide the data destination a list of signals available to be selected for inclusion in a data stream.

4.4.1.2 Stream configuration

This frame serves the same function as the CFG-3 frame from the previous standard. It shall include the information corresponding to the signals being transmitted in a specific data stream identified by its STREAM_ID. This frame shall be transmitted from a data source (PMU or PDC) to the data destination (another PDC).

4.4.2 Device and signal names

The naming convention used for device and signal names included in configuration frames is shown in [Table 2](#). PDC_NAME, PMU_NAME and signal names¹⁸ shall be listed in name fields whose length shall be a number from 1 to 256 B. The first byte in each name field shall be the length, which is an unsigned 8-bit integer that indicates the length of the name in bytes. Subsequent number of bytes shall consist of the name.

A length of 0 indicates no further bytes (no name). In this case, the name field shall consist of a single binary 0. All name fields shall have at least one byte that is the name length. All names shall use UTF-8 coding.

¹⁸Users may wish to consult [\[B16\]](#) for selecting signal names. This standard does not place any requirements on name assignments or their interpretations. It is up to the users (senders and receivers) to assign and interpret names.

Table 2—Name field description

Byte #	Description
First byte (#0)	Name length. Unsigned 8-bit integer.
Remaining bytes (#1 to maximum 255)	Signal or device name. UTF-8 encoding. ^a

^aSome UTF-8 characters require multiple byte combinations.

4.4.3 Configuration frame description

Table 3 shows the details of all configuration frame fields. The stream configuration and capability frames shall include all the fields shown in this table.

Table 3—Configuration frame format

No.	Field	Size (bytes)	Short description
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
7	CONT_IDX	2	Continuation index for fragmented frames. 0: Only fragment in configuration, no further fragments. 1: First fragment in series, more to follow. 2 to 65534: Number of each succeeding fragment, in order. 65535 (hex FFFF): Last fragment in series (see Annex I).
8	TIME_BASE	4	Resolution of FRACSEC time stamp. It shall be an integer multiple of the PMU data transmission rate (see Table 18). It shall also be large enough to support the time resolution required for the discrete event data frames. Bits 31 to 24: Reserved. Set to 0. Bits 23 to 0: 24-bit unsigned integer, which is the subdivision of the second that the FRACSEC is based on.
9	PDC_NAME	1 to 256	PDC name. Could be a name for the composite data stream (see Table 2).
10	NUM_PMU	2	The number of PMUs included in the data frame.
The following fields (11 to 34) are specific to each PMU. They are repeated NUM_PMU times.			
11	PMU_NAME	1 to 256	PMU name—in UTF-8 format with field index (see Table 2).
12	PMU_ID	2	PMU ID number, 16-bit unsigned integer, assigned by user, 1 to 65534 (0 and 65535 are reserved). Identifies the PMU device, which can be physical or virtual. Also see Table 1 for a description of STREAM_ID.
13	PMU_VERSION	2	Bits 15 to 4: Reserved. Set to 0. Bits 3 to 0: Version number, from the SYNC WORD (see Table 1).
14	G_PMU_ID	16	Global PMU ID. Uses RFC 4122, big-endian byte encoding. This is a large enough number to allow a unique identifier to each PMU in the world.
15	FORMAT	2	Data format within the periodic data frame (see Table 4 for details). Bits 15 to 4: Reserved. Set to 0. Bit 3: FREQ/DFREQ. 0 = 16-bit integer, 1 = floating point. Bit 2: ANALOG. 0 = 16-bit integer, 1 = floating point. Bit 1: PHASOR (format). 0 = 16-bit integer, 1 = floating point. Bit 0: PHASOR (encoding). 0 = real and imaginary (rectangular). 1 = magnitude and angle (polar).

Table continues

Table 3—Configuration frame format (*continued*)

No.	Field	Size (bytes)	Short description
16	PHNMR	2	Number of phasors—2-B unsigned integer.
17	ANNMR	2	Number of analog values—2-B unsigned integer.
18	FRNMR	2	Number of frequency signals—2-B unsigned integer.
19	DFDTNMR	2	Number of df/dt signals—2-B unsigned integer.
20	DGNMR	2	Number of digital status words—2-B unsigned integer.
21	CHNAM	1 to 256 per name	Phasor and channel names—In UTF-8 with field index (see Table 2). Minimum of 2 B for each phasor, frequency, ROCOF, analog, and digital channel. Names are in the same order as they are transmitted: all phasors, all frequencies, all ROCOFs, all analogs, and all digitals. For digital channels, the channel name order will be from the least significant to the most significant. For example: the first name is for bit 0 of the first 16-bit status word, the second is for bit 1, etc., up to bit 15. If there is more than 1 digital status, the next name will apply to bit 0 of the second word and so on.

Table continues

Table 3—Configuration frame format (continued)

No.	Field	Size (bytes)	Short description																																										
22	PHSCALE	16 × PHNMR	Conversion factor for phasor channels with flags. Magnitude and angle scaling for phasors with data flags. This factor has four 4-B long words: The first is bit-mapped flags, the second is a magnitude scale factor, the third is an angle offset, and the fourth is voltage class. First 4-B word. First 2 B: 16-bit flag that indicates the type of data modification when data is being modified by a continuous process. When no modification process is being applied, all bits shall be set to 0. (Bit 0 is the LSB, Bit 15 is the MSB) <table><tr><td>Bit #</td><td>meaning when bit set.</td></tr><tr><td>0</td><td>Not used, reserved.</td></tr><tr><td>1</td><td>Up sampled from lower rate.</td></tr><tr><td>2</td><td>Down sampled from higher rate.</td></tr><tr><td>3</td><td>Magnitude filtered.</td></tr><tr><td>4</td><td>Estimated magnitude.</td></tr><tr><td>5</td><td>Estimated angle.</td></tr><tr><td>6</td><td>Phasor magnitude adjusted for calibration.</td></tr><tr><td>7</td><td>Phasor phase adjusted for calibration.</td></tr><tr><td>8</td><td>Phasor phase adjusted for offset (± 30°, ± 120°, etc.).</td></tr><tr><td>9</td><td>Pseudo-phasor value (combined from other phasors).</td></tr><tr><td>10 to 14</td><td>Reserved for future assignment.</td></tr><tr><td>15</td><td>Modification applied. Type not defined here.</td></tr></table> Third byte: phasor type indication (bit 0 is the LSb, bit 7 is the MSb). Bits 07 to 04: Reserved for future use. Set to 0. Bit 03: 0 = voltage; 1 = current. Bits 02 to 00: Phasor component, coded as follows. <table><tr><td>111:</td><td>Reserved.</td></tr><tr><td>110:</td><td>Phase C.</td></tr><tr><td>101:</td><td>Phase B.</td></tr><tr><td>100:</td><td>Phase A.</td></tr><tr><td>011:</td><td>Reserved.</td></tr><tr><td>010:</td><td>Negative sequence.</td></tr><tr><td>001:</td><td>Positive sequence.</td></tr><tr><td>000:</td><td>Zero sequence.</td></tr></table> Fourth byte: available for user designation. Second and third 4-B words. For phasor $X = X_m e^{j\theta}$, this defines the scaling Y and angle offset θ to be applied to the phasor as follows: $X' = Y \times X_m e^{j(\theta-\theta')}$ The second 4-B word is the scale factor Y in 32-bit IEEE floating point. This scales phasor data to primary volts or amperes. If phasors are transmitted in floating-point format and scaled already, this value shall be set to 1.0. The third 4-B word is the phasor angle adjustment θ in radians represented in 32-bit IEEE floating point. If phasors are transmitted in floating-point format and scaled already, this value shall be set to 0.0. The fourth 4-B word is the voltage class for this signal, in 32-bit IEEE floating point format. voltage class shall be used as a general value which approximates the phase-to-phase rms voltage for both the phasor and analog measurements being reported by the PMU. The value is a single precision floating point number in engineering units (e.g., 500 000.0).^a Set to 0.0 if it is not available.	Bit #	meaning when bit set.	0	Not used, reserved.	1	Up sampled from lower rate.	2	Down sampled from higher rate.	3	Magnitude filtered.	4	Estimated magnitude.	5	Estimated angle.	6	Phasor magnitude adjusted for calibration.	7	Phasor phase adjusted for calibration.	8	Phasor phase adjusted for offset (± 30°, ± 120°, etc.).	9	Pseudo-phasor value (combined from other phasors).	10 to 14	Reserved for future assignment.	15	Modification applied. Type not defined here.	111:	Reserved.	110:	Phase C.	101:	Phase B.	100:	Phase A.	011:	Reserved.	010:	Negative sequence.	001:	Positive sequence.	000:	Zero sequence.
Bit #	meaning when bit set.																																												
0	Not used, reserved.																																												
1	Up sampled from lower rate.																																												
2	Down sampled from higher rate.																																												
3	Magnitude filtered.																																												
4	Estimated magnitude.																																												
5	Estimated angle.																																												
6	Phasor magnitude adjusted for calibration.																																												
7	Phasor phase adjusted for calibration.																																												
8	Phasor phase adjusted for offset (± 30°, ± 120°, etc.).																																												
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010:	Negative sequence.																																												
001:	Positive sequence.																																												
000:	Zero sequence.																																												

Table continues

Table 3—Configuration frame format (continued)

No.	Field	Size (bytes)	Short description
23	FRSCALE	8 × FRNMR	Conversion factor for frequency channels. Linear scaling. For value X , this defines scale M and offset B for $X' = M \times X + B$. First 4 B: Magnitude scaling M in 32-bit floating point. Last 4 B: Offset B in 32-bit floating point. If frequency values are transmitted in floating-point format and scaled already, these values shall be set to 1.0, and 0.0, respectively.
24	DFDTSCALE	8 × DFDTNMR	Conversion factor for ROCOF channels. Same as for FRSCALE above.
25	ANSCALE	8 × ANNMR	Conversion factor for analog channels. Same as for FRSCALE above.
26	DIGUNIT	4 × DGNMR	Mask words for digital status words. Two 16-bit words are provided for each digital word. The first will be used to indicate the normal status of the digital inputs by returning a 0 when exclusive ored (XOR) with the status word. The second will indicate the current valid inputs to the PMU by having a bit set in the binary position corresponding to the digital input and all other bits set to 0. ^b
27	PMU_LAT	4	PMU latitude in degrees, WGS84 datum. Range –90.0 to +90.0. Positive values are N of equator. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.
28	PMU_LON	4	PMU longitude in degrees, WGS84 datum. Range –179.99999999 to +180. Positive values are E of the prime meridian. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.
29	PMU_ELEV	4	PMU elevation in meters, WGS84 datum. Positive values are above mean sea level. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.
30	PMUFLAG	2	Bit 15: 1 = PMU does not accept any configuration commands, 0 = PMU accepts configuration commands. Bit 14: 1 = data stream auto starts on power up, 0 = data stream does not auto start on power up. Bit 13: 1 = 50Hz, 0 = 60Hz nominal frequency. Bit 12: 1 = data attributes included in this data stream, 0 = not included Bit 11: 1 = data available for retrieval at this PMU, 0 = not available. ^c Bits 10 to 4: Reserved. Set to 0. Bits 3 to 0: Data filter used. Sixteen possible combinations. 0 = P class. 1 = M class. 2 to 7: reserved. 8 to 15: user defined.
31	WINDOW	4	Phasor measurement window length including all filters and estimation windows in effect. Value is in μ s, 4-B signed integer value (to nearest μ s). A value of –1 indicates the window length is not available. ^d
32	GRP_DLY	4	Phasor measurement group delay including all filters and estimation windows in effect. Value is in μ s, 4-B signed integer value (to nearest μ s). A value of –1 indicates the group delay is not available. ^c
33	PMU_DATA_RATE	2	Rate of data transmissions from this PMU. 2-B integer word (–32 767 to +32 767). For permitted data rates, see 5.1.1. If PMU_DATA_RATE > 0, rate is number of frames per second. If PMU_DATA_RATE < 0, rate is negative of seconds per frame. (e.g., PMU_DATA_RATE = 15 is 15 frames per second) (e.g., PMU_DATA_RATE = –5 is 1 frame per 5 s)
34	CFGCNT	2	Configuration change count. This is incremented each time a change is made in the PMU configuration. 0 is the factory default and the initial value. This count identifies the number of changes in the configuration of this message stream.

Table continues

Table 3—Configuration frame format (*continued*)

No.	Field	Size (bytes)	Short description
Field 11 to field 34 are repeated for as many PMUs as in field 10 (NUM_PMU).			
++	STREAM_DATA_RATE	2	Rate of data transmission for the composite frame in this stream (see PMU_DATA_RATE above for value definition). ^c
++	WAIT_TIME	2	PDC wait time, milliseconds. 2-B unsigned integer.
++	CHK	2	See Table 1.

^aSince this value is part of the configuration frame it can be used to validate measured values and / or used to organize data by voltage class in downstream applications.

^bIf digital status words are used for something other than Boolean status indications (such as minimum or maximum settings), the use of masks is left to the user.

^cThis bit shows if the PMU stores transmitted data. If it does, the receiving PDC may request the stored data by using the old data request command (see 4.6.1.1).

^dFor information only, any required compensation is already applied to the measurement.

^eThe STREAM_DATA_RATE is set independently and need not be the same as the data rates for the included PMUs, which may have different data rates. A PDC may perform data rate conversion.

4.4.4 Fragmentation of configuration frames

If a configuration frame is smaller than 65 535 B, users may use a single frame for its transmission, or may use fragmentation. If a single frame is used, its CONT_IDX value shall be zero. If multiple fragments are used, then regardless of the total frame size, the following shall apply:

- Coordination between fragments of configuration frames shall be achieved by the continuation index (CONT_IDX) field.
- Each fragment shall be 65 535 or fewer bytes.
- Each fragment shall include 16 B of header (field 1 to field 7 in Table 3), with correct CONT_IDX and FRAMESIZE.
- First fragment shall use CONT_IDX of “one.”
- Each subsequent, consecutive fragment shall increment CONT_IDX by one.
- Last fragment shall have CONT_IDX of 65535 (0xFFFF).
- Last two bytes of the last fragment shall be the CHK field for the entire frame. This field shall be calculated with CONT_IDX set to 1, and FRAMESIZE set to 65535.
- Individual fragments shall not include any other CHK field.

The maximum number of fragments are confined by the size of the CONT_IDX field, a 2-B unsigned integer, to 65535. When these frames become larger than 65 535 B, they shall be fragmented for transmission. A method of fragmenting larger frames is described in Annex I.

4.5 Data frames

Two types of data frames are defined: periodic data frames and discrete event data frames. The discrete event data frame is added in this revision. The periodic data frame serves the same function as the data frame in the previous revisions of this standard.

4.5.1 Periodic data frame

A periodic data frame shall contain phasor data measured in real-time and shall be identified by having bits 6 to 4 in the SYNC word set to 000 as shown in Table 1. It shall consist of binary data ordered as shown in Table 4.

All fields shall be fixed length as described and no delimiters shall be used. Measurements shall be performed at a uniform interval and the time stamp included in the frame shall correspond to the measurement time.

Table 4—Periodic data frame format

No.	Field	Size (bytes)	Comment
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
The following fields (7 to 13) are specific to each PMU. They are repeated NUM_PMU times.			
7	STAT_FLAG	2	See Table 5.
8	TIMEQUALITY	2	See Table 6. Also see [B9] for details.
9	PHASOR + (DAPHASOR)	4 or 8 (+2) × PHNMR	Phasor estimates. May be single phase or 3-phase (a, b, c), or symmetrical components (positive, negative, or zero sequence). Values are 4 or 8 B each depending on whether the fixed point 16-bit integer, or floating-point format is used, as indicated by the FORMAT field in configuration frames. The number of values is determined by the PHNMR field in configuration frames (see Table 3). 16-bit integer values: <i>Rectangular format:</i> —Real and imaginary, real value first. —16-bit signed integers, range $-32\,767$ to $+32\,767$. <i>Polar format:</i> —Magnitude and angle, magnitude first. —Magnitude 16-bit unsigned integer range 0 to 65535. —Angle 16-bit signed integer, in radians $\times 10^4$, range $-31\,416$ to $+31\,416$. 32-bit values in IEEE floating-point format: <i>Rectangular format:</i> —Real and imaginary, in engineering units, real value first. <i>Polar format:</i> —Magnitude and angle, magnitude first and in engineering units. —Angle in radians, range $-\pi$ to $+\pi$. Each phasor may be followed by an optional 2-B data attributes field (DAPHASOR). The option to include data attributes in the periodic data frame shall be included in the configuration metadata (see Table 3, Table 8).
10	FREQ + (DAFREQ)	2 or 4 (+2) × FRNMR	Frequency. This may be 16-bit integer or 32-bit floating point, as indicated by the FORMAT field in configuration frames. When fixed point, this field shows frequency deviation from nominal, in mHz. 16-bit signed integers, deviation range $-32\,767$ to $+32\,767$ mHz from nominal (50 Hz or 60 Hz). When 32-bit floating point, this field shows actual frequency value in IEEE floating-point format. Each FREQ measurement may be followed by an optional 2-B data attributes field (DAFREQ). The option to include data attributes in the periodic data frame shall be included in the configuration metadata (see Table 3, Table 8).

Table 4—Periodic data frame format (continued)

No.	Field	Size (bytes)	Comment
11	DFREQ + (DADFREQ)	2 or 4 (+2) x DFDTNMR	ROCOF (fixed or floating point), in Hertz per second times 100. Range -327.67 to $+327.67$ Hz per second. Can be 16-bit integer or IEEE floating point, shall be the same as FREQ above. Data type indicated by the FORMAT field in configuration frames. Each DFREQ measurement may be followed by an optional 2-B data attributes field (DADFREQ). The option to include data attributes in the periodic data frame shall be included in the configuration metadata (see Table 3, Table 8).
12	ANALOG	$2 \times \text{ANNMR}$ or $4 \times \text{ANNMR}$	Analog data. It could be sampled data such as control signal or transducer value. Values and ranges defined by user. 2 or 4 B per value depending on fixed or floating-point format used, as indicated by the FORMAT field in configuration frames. The number of values is determined by the ANNMNR field in configuration frames.
13	DIGITAL	$2 \times \text{DGNMR}$	Digital data, usually representing 16 digital status points (channels). The number of values is determined by the DGNMR field in configuration frames.
Repeat 7 to 13			Field 7 to field 13 are repeated for as many PMUs as in NUM_PMU field in configuration frames.
++	CHK	2	See Table 1.

4.5.1.1 STAT_FLAG word

The STAT_FLAG word field shall be a 16-bit field. Unless specified, each bit shall be set to 0 by the PMU. A PDC, or any other application function may modify the bits only as described in Table 5. Any data described as “PMU only” shall not be modified by any other device or application receiving this data.

Table 5—STAT_FLAG WORD

Bit no.	Set by	Field	Definition
15 to 14	PMU only	Data error	0 0 = Good measurement data, no errors. 1 0 = PMU in test mode. ^a 0 1 = Internal error detected by PMU. Data should be used with caution. 1 1 = Reserved.
13	PMU only	PMU sync ^b	Time sync uncertainty indicator (see 4.5.1.2). 0 = Time sync uncertainty is 250 ns or less. 1 = Time sync uncertainty exceeds 250 ns.
12	PDC	Local time stamp ^c	0 = Normal data. 1 = The data had a different time stamp. The PDC has assigned the current time stamp to this data frame.
11	PDC	Data bad ^d	0 = Normal data. 1 = Do not use. PMU data is included in the data frame, but PDC’s ask users to not use it.
10	PMU/PDC	Inserted data ^e	Always set to 0 by PMU. Set to 1 by PDC only when there is no PMU data, and PDC substitutes it with any data.
9 to 0	PMU	Reserved	Always set to 0 by PMU.

^aThe PMU is operating OK. The data is intended for testing functions, calibration etc. It may or may not be actual power systems measurements.

^bPMU sets this bit, and the TIMEQUALITY field, based on time sync uncertainty. PMUs should continue measuring/streaming irrespective of the uncertainty magnitude, and let users determine data disposition. Originally, this bit was set when the clock was not in sync with a UTC traceable time source, regardless of the magnitude of error/uncertainty.

^cThis bit field is functionally the same as “Sort by Arrival” in IEEE Std C37.118.2-2011 [B7]. It has been renamed in this revision.

^dThis may be due to a known/planned outage. The bit may be set manually at the PDC user interface.

^eThis may be absent data tag, or data calculated by the PDC or an application function. Refer to [B11].

4.5.1.2 Time quality

The time quality field shall be a 16-bit field. Bit usages are described in Table 6 and Table 7. This field shall be set only by the PMU making measurements. Intermediate devices or applications shall not modify any of these fields.

Table 6—TIMEQUALITY field format for version 3 data stream^a

Bit no.	Field	Version 3 data stream
15	Version indicator	0 = Version 3.
14 to 12	Multiplier	10^n .
11 to 0	Time sync uncertainty	Time sync uncertainty in nanoseconds.

^aBits 14 to 0 all being zero means that the estimated uncertainty is less than 0.5 ns, which implies the device is in sync. All being 1 implies the error is (≥ 40.95 s or 4095×10^7 ns), or is unknown. An example: 0x1CE4 or 0001 1100 1110 0100 indicates an error of 33 μ sec (3300×10^1 ns).

There may be occasions when a version 2 and a version 3 data stream are merged to create a version 3 data stream. The adaptation of version 2 data stream time quality attributes into a version 3 data stream time quality field shall be as shown in and Table 7. See Annex H for further details.

Table 7—TIMEQUALITY field format for a version 2 data stream conversion to a version 3 data stream

Bit no.	Field	Conversion from version 2 data stream
15	Version indicator	1 = converted from version 2.
14 to 9	Reserved	Set to 0.
8 to 6	From STAT word	Bits 8 to 6 from STAT word PMU time quality field.
5 to 4	Reserved	Set to 0.
3 to 0	From FRACSEC	Bits 3 to 0 from FRACSEC time quality field.

4.5.1.3 Data attributes

Each phasor, frequency and rate-of-change-of-frequency measurement in the periodic data frame may include an optional 2-B data attributes field, as shown in Table 4, and Table 8. The option shall be set in configuration frames (Table 3, PMUFLAG field), which controls if the DAPHASOR, DAFREQ, and DADFREQ fields are included in a periodic data frame. A PMU output shall include data attributes fields for all its phasors, frequency, and rate-of-change-of-frequency measurements, or none of them.¹⁹ Data attributes fields are not included for analog and digital measurements.

A PDC output with multiple PMUs may include some PMUs with and some without data attributes fields. This is determined by the metadata for each individual PMU. An outside agent (PDC/application/assessment tool) may evaluate the data more thoroughly, based on other factors known to it, and re-assign the data attributes. This shall be done as defined in Table 8. A PDC may add/remove/modify data attributes fields in its output stream, regardless of what the PMU may send. A PDC shall modify the configuration metadata for its output stream to indicate the changes made.

The individual data attributes bits for all data attribute fields are defined in Table 8. They shall be set to 1 only as described in Table 8. They shall be set to 0 by the PMU in all other cases.

¹⁹Previous version of this standard had a common data quality field for all measurements from a PMU.

Table 8—Data attributes bit definitions

Bit #	Field	Set By	Description
15	Data modified	PDC	1 = Modified by PDC/application function.
14	Data error	PMU	1 = Erroneous data. Do not use.
13	Test data	PMU	1 = Test data. Data from PMU, but not actual system measurement. Not to be used for application functions, like test mode in STAT_FLAG word (Table 5). The data shall not be rejected by the PDC but shall be passed through as is.
12	Step detected	PMU	1 = Step/fault detected. Measurement window encompasses a transient.
11	Repeated value	PMU/PDC	1 = Last known value – repeated data.
10	Inconsistent data	PDC	1 = Inconsistent data, based on nearby observations. Use with caution, value is suspect.
9	Estimated value	PDC	1 = Estimated value.
8	Phase adjusted	PDC	1 = Phase adjusted for reference phase rotation in system.
7	Format change type	PDC	1 = Format change (int/float).
6	Format change type	PDC	1 = Format change (polar/rectangular).
5	Reserved	PMU	Always set to 0 by PMU.
4	Reserved	PMU	Always set to 0 by PMU.
3	Value unreasonable	PDC/ application	1 = Engineering values not within reasonable range.
2	Reserved	PMU	Always set to 0 by PMU.
1	Reserved	PMU	Always set to 0 by PMU.
0	Reserved	PMU	Always set to 0 by PMU.

4.5.2 Discrete event data frame

A discrete event data frame shall contain digital data measured in real-time and shall be identified by having bits 7 to 4 in the SYNC word set to 1001 as shown in Table 1. It shall consist of binary data ordered as shown in Table 9. All fields shall be fixed length as described and no delimiters shall be used. The time stamp included in the frame shall correspond to the nearest instant that can be expressed using the user specified fractional time interval (see 4.3.1.4), for the detected change. These may or may not match the uniform time stamps associated with the periodic data frames.

Table 9—Discrete events data frame format—all digitals^a

No.	Field	Size (bytes)	Description
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
7	DIGITAL	2 × DGNMR	All digital data, in the same sequence as in the periodic data frame.
8+	Repeat 7	Repeated for NUM_PMU.	
++	CHK	2	See Table 1.

^aThe discrete event data frame includes all digital signals, and only the digital signals, from the periodic data frame.

4.6 Commands²⁰

A data sending device (PMU or PDC) shall be able to receive commands and take appropriate actions. These commands are described in the following sections.

4.6.1 Command frame

The command frame is identified by bits 7 to 4 of the SYNC word as shown in Table 1. It shall use the same SYNC, FRAMESIZE, STREAM_ID, SOC, LEAP_BYTE, FRACSEC, and CHK words as all other messages as shown in Table 1. The command frame is shown in Table 10. The PMU/PDC shall match the STREAM_ID with a valid code stored internally before accepting and executing the command. The PMU/PDC shall match the command code with that of a configured output stream and execute the command for the indicated data stream output, leaving other streams, if there are any, unchanged. CMD shall be a 2-B command code as defined in Table 11.

Table 10—Command frame format

No.	Field	Size (bytes)	Definition
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
7	CMD	2	Command being sent to the PMU/PDC (see Table 11).
8	EXTFRAME	0 to 65518	Extended frame data, 16-bit words, 0 B to 65518 B as indicated by frame size, data user defined.
9	CHK	2	See Table 1.

Table 11—Commands sent to the PMU/PDC

Command word bits	Description
15 to 0:	—
0000 0000 0001 0000	Turn off transmission of data frames.
0000 0000 0010 0000	Turn on transmission of data frames.
0000 0000 0011 0000	Reserved.
0000 0000 0100 0000	Send capability.
0000 0000 0101 0000	Reserved.
0000 0000 0110 0000	Send stream configuration frame.
0000 0000 1000 0000	Extended frame. Old data request (optional).
0000 0000 xxxx xxxx	All undesignated codes reserved.
0000 yyyy xxxx xxxx	All codes where yyyy ≠ 0 available for user designation.
zzzz xxxx xxxx xxxx	All codes where zzzz ≠ 0 reserved.

²⁰Time stamps are included for uniformity between all synchrophasor frames. Their values in command frames generally do not get used unless the user wishes to keep an archive for reference. They need not have the accuracy needed by data frames.

4.6.1.1 Old data request (optional)

Data frames may get lost occasionally due to communications problems. If the PMU or intervening PDC does not store such data, the data loss is irretrievable. In cases where data is stored, the data may still be retrieved and used for post-mortem analysis and archiving, though probably not in time for protection and control functions. It may be possible to retrieve such stored data through physical access, but this new protocol feature allows retrieving it directly and automatically as part of the data reporting function.

The data destination (such as the control center) shall request such data, specifying stream id and time stamp range by using a command frame as shown in Table 12. If the data is available to be streamed in the format as implemented at the time of this request, the source (such as the PDC) shall respond with data frames (both periodic and discrete event) in the same stream, with the correct time stamps for the data being transmitted. The receiver shall accept such data with past time stamps.

Table 12—Old data request command

No.	Field	Size (bytes)	Definition		
1	SYNC	2	See Table 1.		
2	FRAMESIZE	2	See Table 1.		
3	STREAM_ID	2	See Table 1.		
4	SOC	4	See Table 1.		
5	LEAP_BYTE	1	See Table 1.		
6	FRACSEC	3	See Table 1.		
7	CMD	2	0000 0000 1000 0000 – Extended frame		
8	EXTFRAME	18	Old data request	2	0000 0000 0000 0001
			Data start time	4	SOC 1
				4	FRACSEC 1
			Data stop time	4	SOC 2
				4	FRACSEC 2
9	CHK	2	See Table 1.		

These data transmissions shall be interspersed with the existing data stream. The sender may select the number of extra frames to be sent between normal data transmissions, depending on bandwidth availability.

If the data is not available, or only partially available, the source shall respond in accordance with an error-response described in 4.7.

NOTE—This command is not available in previous revisions of this standard.

4.6.1.2 Stream ID available (optional)

The data destination (such as the control center) shall request the data source, if the specified STREAM_ID is available to be set up as a new stream, or if it is in use already, or if the source cannot set up any new streams, by using a command frame as shown in Table 13. The data source shall respond by indicating it is available, or it is not available with the remaining number of streams available (including zero, if no new streams can be set up), or command rejected. These error-responses are detailed in 4.7.

Table 13—Stream ID available command

No.	Field	Size (bytes)	Definition		
1	SYNC	2	See Table 1.		
2	FRAMESIZE	2	See Table 1.		
3	STREAM_ID	2	See Table 1.		
4	SOC	4	See Table 1.		
5	LEAP_BYTE	1	See Table 1.		
6	FRACSEC	3	See Table 1.		
7	CMD	2	0000 0000 1000 0000 – Extended frame.		
8	EXTFRAME	4	Stream ID available?	2	0000 0000 0000 0010
			Stream ID	2	Valid stream ID (see Table 1).
9	CHK	2	See Table 1.		

4.6.2 Configuration commands (optional)

This revision of the standard adds commands to allow remote configuration of data streams, from a data destination to a data source (such as a control center to a substation PDC, or a substation PDC to a PMU). These commands are optional. An example of a configuration method is shown in Annex E.

4.6.2.1 Rename signals

The rename signals frame is identified by Bits 7 to 4 of the SYNC word as shown in Table 1. This frame serves as a command from the data destination to the data source, to change signal names and other identifiers for the stream identified by the STREAM_ID. The rename frame shall include data as shown in Table 14. This is a subset of configuration frames (see Table 3). This command shall not be accepted by the data source if the data stream is ON. An error-response shall be sent if the command is rejected for any reason (see 4.7).

The entire frame shall be sent, even if only one name or identifier is to be changed. The receiver shall replace all previous values with the new values sent.²¹ The fields indicating a number (NUM_PMU, PHNMNR, ANNMNR, FREQNMR, DFDTNMR) shall match the corresponding numbers of that STREAM_ID.

Table 14—Rename signals frame format^a

No.	Field	Size (bytes)	Short description
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
7	CONT_IDX	2	See Table 3.
8	PDC_NAME	1 to 256	New name to be assigned.
9	NUM_PMU	2	Shall match the original value.
The following fields (10 to 21) are specific to each PMU. They are repeated NUM_PMU times.			
10	PMU_NAME	1 to 256	New name to be assigned.
11	PMU_ID	2	New value to be assigned.
12	G_PMU_ID	16	New value to be assigned.
13	PHNMNR	2	Shall match the original value.

Table continues

²¹If any of the values are to be retained, the same values should be sent.

Table 14—Rename signals frame format^a (continued)

No.	Field	Size (bytes)	Short description
14	ANNMR	2	Shall match the original value.
15	FRNMR	2	Shall match the original value.
16	DFDTNMR	2	Shall match the original value.
17	DGNMR	2	Shall match the original value.
18	CHNAM	1 to 256 per name	New names to be assigned.
19	PMU_LAT	4	New value to be assigned.
20	PMU_LON	4	New value to be assigned.
21	PMU_ELEV	4	New value to be assigned.
<i>Repeat 10 to 21</i>			Field 10 to field 21, repeated for as many PMUs as in (NUM_PMU) field.
++	CHK	2	See Table 1.

^aThese commands are intended for use during initial set up, or for any reconfiguration during normal operation. They are not intended for use for dynamic configuration changes.

4.6.2.2 Configure stream

The configure stream frame is identified by bits 7 to 4 of the SYNC word as shown in Table 1. This frame serves as a command from the data destination to the data source, to set up a new data stream. If the STREAM_ID is not in use, a new data stream shall be set up in accordance with this command. If the STREAM_ID is in use by an existing data stream, that stream shall be replaced by a new stream, in accordance with this command. This frame shall include data as shown in Table 15. This is a subset of configuration frames (see Table 3). This command shall not be accepted by the data source if the data stream is ON. An error-response shall be sent if the command is rejected for any reason (see 4.7).

The fields indicating any identity (PDC_NAME, PMU_NAME, PMU_ID, G_PMU_ID, CHNAM) shall match the values sent originally by the data source for the STREAM_ID.

Table 15—Configure stream frame format^a

No.	Field	Size (bytes)	Short description
1	SYNC	2	—
2	FRAMESIZE	2	—
3	STREAM_ID	2	New STREAM_ID to set up a new data stream. Existing STREAM_ID overwrites the existing data stream.
4	SOC	4	—
5	LEAP_BYTE	1	—
6	FRACSEC	3	—
7	CONT_IDX	2	—
8	TIME_BASE	4	New value to be assigned.
9	PDC_NAME	1 to 256	Shall match the original value.
10	NUM_PMU	2	New value to be assigned.
The following fields (11 to 30) are specific to each PMU. They are repeated NUM_PMU times.			
11	PMU_NAME	1 to 256	Shall match the original value.
12	PMU_ID	2	Shall match the original value.
13	G_PMU_ID	16	Shall match the original value.
14	FORMAT	2	New value to be assigned.
15	PHNMR	2	New value to be assigned.

Table continues

Table 15—Configure stream frame format^a (continued)

No.	Field	Size (bytes)	Short description
16	ANNMR	2	New value to be assigned.
17	FRNMR	2	New value to be assigned.
18	DFDTNMR	2	New value to be assigned.
19	DGNMR	2	New value to be assigned.
20	CHNAM	1 to 256 per name	Shall match the current values.
21	PHSCALE	16 × PHNMR	New value to be assigned.
22	FRSCALE	8 × FRNMR	New value to be assigned.
23	DFDTSIZE	8 × DFDTNMR	New value to be assigned.
24	ANSIZE	8 × ANNMR	New value to be assigned.
25	DIGUNIT	4 × DGNMR	New value to be assigned.
26	PMUFLAG	2	New value to be assigned.
27	WINDOW	4	New value to be assigned.
28	GRP_DLY	4	New value to be assigned.
29	PMU_DATA_RATE	2	New value to be assigned.
30	CFGCNT	2	New value to be assigned.
<i>Repeat 11 to 30</i>			Fields 11 to 30, repeated for as many PMUs as in (NUM_PMU) field.
++	STREAM_DATA_RATE	2	New value to be assigned.
++	WAIT_TIME	2	New value to be assigned.
++	CHK	2	See Table 1.

^a These commands are intended for use during initial set up, or for any reconfiguration during normal operation. They are not intended for use for dynamic configuration changes.

4.7 Error-response frame

At times, a data sending device (PMU or PDC) may receive an invalid command, or a command it cannot respond to. In such instances, the data sending device shall respond with an error-response frame as shown in Table 16. This frame is identified by bits 7 to 4 of the SYNC word as shown in Table 1. It uses the same SYNC, FRAMESIZE, STREAM_ID, SOC, LEAP_BYTE, FRACSEC, and CHK words as all other messages as shown in Table 1. The error-response Word is shown in Table 17. PMUs that are configured to not accept commands (See PMUFLAG, bit 15 in Table 3) are not required to send error-response frames. However, they shall send error-response frames when configured to do so, as may happen during testing.

Table 16—Error-response frame description

No.	Field	Size (bytes)	Comment
1	SYNC	2	See Table 1.
2	FRAMESIZE	2	See Table 1.
3	STREAM_ID	2	See Table 1.
4	SOC	4	See Table 1.
5	LEAP_BYTE	1	See Table 1.
6	FRACSEC	3	See Table 1.
7	ERROR-RESPONSE-1	2	Error-response word 1 defined in Table 17.
8	ERROR-RESPONSE-2	2	Error-response word 2. It shall be 0, unless defined otherwise in Table 17.
9	CHK	2	See Table 1.

Table 17—Error-response words 1 and 2

ERROR-RESPONSE-1 decimal value	Description
0	Reserved.
1	Could not interpret command or rejected command.
2	Wrong STREAM_ID or PMU_ID. Users may enable this during testing/calibration/installation etc. and disable it during normal operation.
3 to 31	Reserved.
Rename command errors	
32	Number mismatch. One of the (NUM_PMU, PHNMNR, ANNMR, FREQNMR, DFDTNMR) does not match. ERROR-RESPONSE-2 indicates the first of the expected matching fields with the mismatch.
33	Too many names included. ERROR-RESPONSE-2 indicates which PMU number has this error.
34	Too few names included. ERROR-RESPONSE-2 indicates which PMU number has this error.
35	Unacceptable characters in a name (see Table 2). ERROR-RESPONSE-2 indicates first of the expected names with this error.
36	Data stream is ON. Command not accepted.
37 to 63	Reserved.
Configure stream command errors. In most cases, ERROR-RESPONSE-2 indicates the sequence number of the field where the error occurred.	
64	Invalid TIME_BASE. The assigned value does not support any of the data rates in Table 18.
65	Mismatched field. ERROR-RESPONSE-2 indicates which expected matching field does not match.
66	NUM_PMU error. Too many PMUs, or zero PMUs. ERROR-RESPONSE-2 indicates the number.
67	FORMAT field invalid.
68	Number mismatch. One of the (PHNMNR, ANNMR, FREQNMR, DFDTNMR) is larger than the number of signals available. ERROR-RESPONSE-2 indicates the first of the expected matching fields with the mismatch.
69	PHSCALE invalid. ERROR-RESPONSE-2 indicates the first of the fields with this error.
70	Floating point number error in FRSCALE, DFDTSKALE, ANSCALE. ERROR-RESPONSE-2 indicates the first of the fields with this error.
71	Invalid LATITUDE. ERROR-RESPONSE-2 indicates which PMU number has this error.
72	Invalid LONGITUDE. ERROR-RESPONSE-2 indicates which PMU number has this error.
73	Invalid PMUFLAG. ERROR-RESPONSE-2 indicates which PMU number has this error.
74	Invalid WINDOW (a negative number, less than −1). ERROR-RESPONSE-2 indicates which PMU number has this error.
75	Invalid GRP_DLY (a negative number, less than −1). ERROR-RESPONSE-2 indicates which PMU number has this error.
76	Invalid PMU_DATA_RATE. ERROR-RESPONSE-2 indicates which PMU number has this error.

Table continues

Table 17—Error-response words 1 and 2 (continued)

Configure stream command errors. In most cases, ERROR-RESPONSE-2 indicates the sequence number of the field where the error occurred.	
77	Invalid STREAM_DATA_RATE. ERROR-RESPONSE-2 indicates the value.
78	Data stream is ON. Command not accepted.
79 to 95	Reserved.
Old data request commands	
96	No requested data available.
97	Partial requested data available.
98 to 127	Reserved.
Stream ID available commands	
128	Stream ID available.
129	Stream ID not available. ERROR-RESPONSE-2 indicates the number of stream IDs available for use.
Other commands	
130 to 1023	Reserved.
1024 to 61439	Available for user designation.
61440 to 65535	Reserved.

5. Communications

This standard defines a message structure for use with real-time data transmission, where data is sent immediately after measurement. The data transmission is periodic or discrete event based. This standard does not specify any communication channel or method. Any communication system with adequate bandwidth may be used.

5.1 Periodic data transmission

Synchrophasor estimates shall be made so they can be reported (transmitted) as periodic data frames at a rate F_s that is an integer number of frames per second or integer number of seconds per frame as specified by the PMU_DATA_RATE and STREAM_DATA_RATE variables in the stream configuration frame (Table 3). A periodic data frame is a set of measurements that may include multiple channels of phasor estimates, optional data attributes words, analog words, and digital words with a measurement status word and a time stamp as defined in 4.3.1. Depending on the data length, a frame may be segmented into several internet protocol (IP) packets.

5.1.1 Data reporting rates

For periodic data frames, PMUs shall support data reporting (by recording or output) at sub-multiples of the nominal power system frequency. Required rates for 50 Hz and 60 Hz systems are listed in Table 18. Any device or system complying with this standard shall support at least one of the rates listed in Table 18 for 50 Hz and/or 60 Hz systems.

Table 18—Required PMU reporting rates

System frequency	50 Hz			60 Hz					
Reporting rates (F_s —frames per second)	10	25	50	10	12	15	20	30	60

The actual rate to be used shall be user selectable. Support for other reporting rates is permissible, and higher rates such as 100 frames/s or 120 frames/s and rates lower than 10 frames/s such as 1 frame/s are allowed.

5.1.2 Reporting times

For a reporting rate, N frames per second where N is a positive integer, the reporting times shall be evenly spaced through each second with the time of the first frame coincident with the UTC second rollover. Thus, the first frame shall be frame number 0 (frames numbered 0 thru $N - 1$) with a FRACSEC time of 0, the next frame number 1 with a fractional time $1 / N$, and so on through frame $N - 1$ with a fractional time $(N - 1) / N$. These reporting times (time stamps) are to be used for determining the instantaneous values of the synchrophasor as defined in B.2. If rates lower than 1/s are used, there shall be one report on the hour (xx:00:00) and evenly spaced reports thereafter according to the chosen rate. (See TIME_BASE in Table 3, FRACSEC in Table 1 and TIMESTAMP in 4.3.1.4).

5.2 Discrete event frames transmission

Discrete event frames are transmitted immediately following the event. The reporting rate feature does not apply to them. FRACSEC may have any integer value less than TIME_BASE. This allows for a greater precision and accuracy to their reporting times, compared to the periodic data frame.

Annex A

(normative)

Version 2 compliance requirements (optional)

A.1 Introduction

The 2011 edition of this standard [B7] describes the version 2 of this standard. This annex summarizes those requirements. These requirements shall be met by devices and applications to remain compliant with version 2 of this standard.

A.2 Message framework

Four message types are defined here: *data*, *configuration*, *header*, and *command*. The first three message types are transmitted from the PMU/PDC that serves as the data source, and the last (command) is received by the PMU/PDC. All messages are transmitted as shown in Figure A.1.

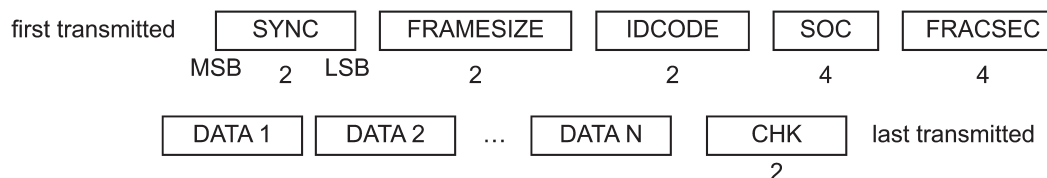


Figure A.1—Example of frame transmission order for version 2

Table A.1—Word definitions common to all frame types for version 2

Field	Size (bytes)	Comments
SYNC	2	Frame synchronization word. Leading byte: AA hex. Second byte: Frame type and version, divided as follows: Bit 7: Reserved for future definition, must be 0 for standard version 2. Bits 6 to 4: For version 2 (shown in this Annex). 000: data frame. 001: header frame. 010: configuration frame 1. 011: configuration frame 2. 101: configuration frame 3. 100: command frame (received message). Bits 3 to 0: Version number, in binary (1 to 15) Version 1 (0001) for messages defined in IEEE Std C37.118-2005 [B5]. Version 2 (0010) for messages defined in IEEE Std C37.118.2-2011 [B7].
FRAMESIZE	2	Total number of bytes in the frame, including CHK. 16-bit unsigned number. Range = maximum 65535.
IDCODE	2	Data stream ID number, 16-bit integer, assigned by user, 1 to 65534 (0 and 65535 are reserved). Identifies destination data stream for commands and source data stream for other messages. A stream will be hosted by a device that can be physical or virtual. If a device only hosts one data stream, the IDCODE identifies the device as well as the stream. If the device hosts more than one data stream, there shall be a different IDCODE for each stream.

Table continues

Table A.1—Word definitions common to all frame types for version 2 (continued)

Field	Size (bytes)	Comments
SOC	4	Time stamp, 32-bit unsigned number, SOC count starting at midnight 01-Jan-1970 (UNIX time base). Range is 136 yrs, rolls over 2106 AD. Leap seconds are not included in count, so each year has the same number of seconds except leap years, which have an extra day (86 400 s).
FRACSEC	4	Fraction of second and time quality, time of measurement for data frames or time of frame transmission for non-data frames. Bits 31 to 24: Message time quality as defined in A.2.1. Bits 23 to 00: FRACSEC, 24-bit integer number. When divided by TIME_BASE yields the actual fractional second. FRACSEC used in all messages to and from a given PMU shall use the same TIME_BASE that is provided in the configuration message from that PMU.
CHK	2	CRC-CCITT, 16-bit unsigned integer.

A.2.1 Time and message quality

The 32-bit (4-B) FRACSEC is divided into two components: a 24-bit unsigned integer that is the actual FRACSEC count and an 8-bit message time quality flag. Time calculation is like the one shown in 4.3.1.4.

Table A.2—Time quality flag bit definitions for version 2

Bit #	Description
7	Reserved.
6	Leap second direction—0 for add, 1 for delete.
5	Leap second occurred—set in the first second after the leap second occurs and remains set for 24 h.
4	Leap second pending—shall be set not more than 60 s nor less than 1 s before a leap second occurs, and cleared in the second after the leap second occurs.
3 to 0	Message time quality indicator code—see Table A.3.

The message time quality indicator code contained in the lowest 4 bits indicates the maximum time error as determined by the PMU/PDC clock function and shown in Table A.3.

Table A.3—4-bit message time quality indication codes (MSG_TQ) for version 2

BINARY	HEX	Value (worst-case accuracy)
1111	F	Fault—clock failure, time not reliable.
1011	B	Time within 10 s of UTC.
1010	A	Time within 1 s of UTC.
1001	9	Time within 10^{-1} s of UTC.
1000	8	Time within 10^{-2} s of UTC.
0111	7	Time within 10^{-3} s of UTC.
0110	6	Time within 10^{-4} s of UTC.
0101	5	Time within 10^{-5} s of UTC.
0100	4	Time within 10^{-6} s of UTC.
0011	3	Time within 10^{-7} s of UTC.
0010	2	Time within 10^{-8} s of UTC.
0001	1	Time within 10^{-9} s of UTC.
0000	0	Normal operation, clock locked to UTC traceable source.

A.3 Data frame

A data frame contains measured data. Data frame details are shown in [Table A.4](#) and [Table A.5](#).

Table A.4—Data frame organization for version 2

No.	Field	Size (bytes)	Comment
1	SYNC	2	Sync byte followed by frame type and version number.
2	FRAMESIZE	2	Number of bytes in frame, defined in A.2 .
3	IDCODE	2	Stream source ID number, 16-bit integer, defined in A.2 .
4	SOC	4	SOC time stamp, defined in A.2 , for all measurements in frame.
5	FRACSEC	4	Fraction of second and time quality, defined in A.2 , for all measurements in frame.
6	STAT	2	Bit-mapped flags.
7	PHASORS	4 × PHNMR or 8 × PHNMR	Phasor estimates. May be single phase or 3-phase positive, negative, or zero sequence. Four or 8 B each depending on the fixed 16-bit or floating-point format used, as indicated by the FORMAT field in the configuration frame. The number of values is determined by the PHNMR field in configuration 1, 2, and 3 frames.
8	FREQ	2 / 4	Frequency (fixed or floating point).
9	DFREQ	2 / 4	ROCOF (fixed or floating point).
10	ANALOG	2 × ANNMR or 4 × ANNMR	Analog data, 2 B or 4 B per value depending on fixed or floating-point format used, as indicated by the FORMAT field in configuration 1, 2, and 3 frames. The number of values is determined by the ANNMR field in configuration 1, 2, and 3 frames.
11	DIGITAL	2 × DGNMR	Digital data, usually representing 16 digital status points (channels). The number of values is determined by the DGNMR field in configuration 1, 2, and 3 frames.
<i>Repeat 6 to 11</i>			Field 6 to field 11 are repeated for as many PMUs as in NUM_PMU field in configuration frame.
12+	CHK	2	CRC-CCITT.

Table A.5—Word definitions unique to data frames for version 2

Field	Size (bytes)	Comments
SYNC	2	First byte: AA hex. Second byte: 01 hex (frame is version 1, IEEE Std C37.118-2005 [B5]).

Table continues

Table A.5—Word definitions unique to data frames for version 2 (continued)

Field	Size (bytes)	Comments	
STAT	2	Bit mapped flags. Bits 15 to 14: Data error: 00 = good measurement data, no errors. 01 = PMU error. No information about data. 10 = PMU in test mode (do not use values) or <i>absent data tags</i> have been inserted (do not use values). 11 = PMU error (do not use values). Bit 13: PMU sync, 0 when in sync with a UTC traceable time source. Bit 12: Data sorting, 0 by time stamp, 1 by arrival. Bit 11: PMU trigger detected, 0 when no trigger. Bit 10: Configuration change, set to 1 for 1 min to advise configuration will change, and clear to 0 when change effected. Bit 09: Data modified, 1 if data modified by post processing, 0 otherwise. Bits 08 to 06: PMU time quality. Refer to codes in Table A.6 . Bits 05 to 04: Unlocked time: 00 = sync locked or unlocked < 10 s (best quality) 01 = 10 s ≤ unlocked time < 100 s 10 = 100 s < unlock time ≤ 1000 s 11 = unlocked time > 1000 s Bits 03 to 00: Trigger reason: 1111 to 1000: Available for user definition	
		0111: Digital 0110: Reserved 0101: df/dt High 0100: Frequency high or low	0011: Phase angle diff 0010: Magnitude high 0001: Magnitude low 0000: Manual
PHASORS	4 / 8	Data type indicated by the FORMAT field in configuration 1, 2, and 3 frames. 16-bit integer values: <i>Rectangular format:</i> —Real and imaginary, real value first. —16-bit signed integers, range −32 767 to +32 767. <i>Polar format:</i> —Magnitude and angle, magnitude first. —Magnitude 16-bit unsigned integer range 0 to 65 535. —Angle 16-bit signed integer, in radians × 10 ⁴ , range −31 416 to +31 416. 32-bit values in IEEE floating-point format: <i>Rectangular format:</i> —Real and imaginary, in engineering units, real value first. <i>Polar format:</i> —Magnitude and angle, magnitude first and in engineering units. —Angle in radians, range −π to +π.	
FREQ	2 / 4	Frequency deviation from nominal, in mHz. Range—nominal (50 Hz or 60 Hz) −32.767 to +32.767 Hz. 16-bit integer or 32-bit floating point. 16-bit integer: 16-bit signed integers, range −32 767 to +32 767. 32-bit floating point: Actual frequency value in IEEE floating-point format. Data type indicated by the FORMAT field in configuration 1, 2, and 3 frames.	
DFREQ	2 / 4	ROCOF, in hertz per second times 100. Range −327.67 to +327.67 Hz per second. Can be 16-bit integer or IEEE floating point, same as FREQ above. Data type indicated by the FORMAT field in configuration 1, 2, and 3 frames.	
ANALOG	2 / 4	Analog word. 16-bit integer. It could be sampled data such as control signal or transducer value. Values and ranges defined by user. Can be 16-bit integer or IEEE floating point. Data type indicated by the FORMAT field in configuration 1, 2, and 3 frames.	
DIGITAL	2	Digital status word. It could be bit mapped status or flag. Values and ranges defined by user.	

A.3.1 PMU time quality codes in STAT word

These are shown in [Table A.6](#).

Table A.6—3-bit PMU time quality indication codes (PMU_TQ) for version 2

BINARY	HEX	VALUE (worst-case accuracy)
111	7	Estimated maximum time error > 10 ms or time error unknown.
110	6	Estimated maximum time error < 10 ms.
101	5	Estimated maximum time error < 1 ms.
100	4	Estimated maximum time error < 100 μs.
011	3	Estimated maximum time error < 10 μs.
010	2	Estimated maximum time error < 1 μs.
001	1	Estimated maximum time error < 100 ns.
000	0	Not used (indicates code from previous version of profile).

A.4 Configuration frame

A configuration frame contains information and processing parameters for a synchrophasor data stream. Three types of configuration frames are defined. CFG-1, CFG-2, and an optional CFG-3. CFG-1 and CFG-2 details are shown in [Table A.7](#) and [Table A.8](#). CFG-1 denotes the PMU/PDC capability, indicating all the data that the PMU/PDC is capable of reporting. CFG-2 indicates measurements currently being reported (transmitted) in the data frame.

Table A.7—Configuration frame 1 and 2 organization, for version 2

No	Field	Size (bytes)	Short description
1	SYNC	2	Sync byte followed by frame type and version number.
2	FRAMESIZE	2	Number of bytes in frame, defined in A.2 .
3	IDCODE	2	Stream source ID number, 16-bit integer, defined in A.2 .
4	SOC	4	SOC time stamp, defined in A.2 .
5	FRACSEC	4	Fraction of second and message time quality, defined in A.2 .
6	TIME_BASE	4	Resolution of FRACSEC time stamp.
7	NUM_PMU	2	The number of PMUs included in the data frame.
8	STN	16	Station name—16 B in ASCII format.
9	IDCODE	2	Data source ID number identifies source of each data block.
10	FORMAT	2	Data format within the data frame.
11	PHNMR	2	Number of phasors—2-B integer (0 to 32 767).
12	ANNMR	2	Number of analog values—2-B integer.
13	DGNMR	2	Number of digital status words—2-B integer.
14	CHNAM	16 × (PHNMR + ANNMNR + 16 × DGNMR)	Phasor and channel names—16 B for each phasor, analog, and each digital channel (16 channels in each digital word) in ASCII format in the same order as they are transmitted. For digital channels, the channel name order will be from the least significant to the most significant (e.g., the first name is for bit 0 of the first 16-bit status word, the second is for bit 1, etc., up to bit 15). If there is more than 1 digital status, the next name will apply to bit 0 of the second word and so on.
15	PHUNIT	4 × PHNMR	Conversion factor for phasor channels.
16	ANUNIT	4 × ANNMNR	Conversion factor for analog channels.
17	DIGUNIT	4 × DGNMR	Mask words for digital status words.

Table continues

Table A.7—Configuration frame 1 and 2 organization, for version 2 (continued)

No	Field	Size (bytes)	Short description
18	FNOM	2	Nominal line frequency code and flags.
19	CFGCNT	2	Configuration change count.
Repeat 8 to 19			Field 8 to field 19, repeated for as many PMUs as in field 7 (NUM_PMU).
20+	DATA_RATE	2	Rate of data transmissions.
21+	CHK	2	CRC-CCITT.

Table A.8—Word definitions unique to configuration frames 1 and 2, for version 2

Field	Size (bytes)	Description
SYNC	2	First byte: AA hex Second byte: 21 hex for configuration 1 31 hex for configuration 2 Both frames are version 1 (IEEE Std C37.118-2005 [B5]).
TIME_BASE	4	Resolution of the fractional second time stamp (FRACSEC) in all frames. Bits 31 to 4: Reserved for flags (high 8 bits) Bits 23 to 0: 24-bit unsigned integer, which is the subdivision of the second that the FRACSEC is based on The actual “fractional second of the data frame” = FRACSEC/TIME_BASE.
NUM_PMU	2	The number of PMUs included in the data frame. No limit specified. The actual limit will be determined by the limit of 65 535 B in one frame (“FRAMESIZE” field).
STN	16	Station name—16 B in ASCII format.
IDCODE	2	Data stream ID number, 16-bit integer, defined in A.2. It identifies the data stream in field 3 and the data source in fields 9 and higher. Field 3 identifies the stream that is being received. The IDCODEs in field 9 (and higher if more than one PMU data is present) identify the original source of the data and will usually be associated with a particular PMU. The IDCODEs in a data stream received directly from a PMU will usually be the same.
FORMAT	2	Data format in data frames, 16-bit flag. Bits 15 to 4: Unused Bit 3: 0 = FREQ/DFREQ 16-bit integer, 1 = floating point Bit 2: 0 = analogs 16-bit integer, 1 = floating point Bit 1: 0 = phasors 16-bit integer, 1 = floating point Bit 0: 0 = phasor real and imaginary (rectangular), 1 = magnitude and angle (polar)
PHNMR	2	Number of phasors—2-B integer.
ANNMR	2	Number of analog values—2-B integer.
DGNMR	2	Number of digital status words—2-B integer. Digital status words are normally 16-bit Boolean numbers with each bit representing a digital status channel measured by a PMU. A digital status word may be used in other user-designated ways.
CHNAM	16	Phasor and channel names—16 B for each phasor, analog and digital status word in ASCII format in the same order as they are transmitted.
PHUNIT	4	Conversion factor for phasor channels. Four bytes for each phasor. Most significant byte: 0 = voltage; 1 = current. Least significant bytes: An unsigned 24-bit word in 10 ⁻⁵ V or amperes per bit to scale 16-bit integer data (if transmitted data is in floating-point format, this 24-bit value shall be ignored).
ANUNIT	4	Conversion factor for analog channels. Four bytes for each analog value. Most significant byte: 0—single point-on-wave, 1—rms of analog input, 2—peak of analog input, 5 to 64—reserved for future definition; 65 to 255—user definable. Least significant bytes: A signed 24-bit word, user defined scaling.

Table continues

Table A.8—Word definitions unique to configuration frames 1 and 2, for version 2 (continued)

Field	Size (bytes)	Description
DIGUNIT	4	Mask words for digital status words. Two 16-bit words are provided for each digital word. The first will be used to indicate the normal status of the digital inputs by returning a 0 when exclusive ored (XOR) with the status word. The second will indicate the current valid inputs to the PMU by having a bit set in the binary position corresponding to the digital input and all other bits set to 0. See NOTE.
FNOM	2	Nominal line frequency code (16-bit unsigned integer). Bits 15 to 1: Reserved Bit 0: 1—Fundamental frequency = 50 Hz 0—Fundamental frequency = 60 Hz
DATA_RATE	2	Rate of phasor data transmissions—2-B integer word (–32 767 to +32 767) If DATA_RATE > 0, rate is number of frames per second. If DATA_RATE < 0, rate is negative of seconds per frame. E.g., DATA_RATE = 15 is 15 frames per second; DATA_RATE = –5 is 1 frame per 5 s.
CFGCNT	2	Configuration change count is incremented each time a change is made in the PMU configuration. 0 is the factory default and the initial value. This count identifies the number of changes in the configuration of this message stream. The count will be the same in all configuration messages (CFG-1, CFG-2, and CFG-3) for the same message revision.

NOTE—If digital status words are used for something other than Boolean status indications, the use of masks is left to the user, such as min or max settings.

CFG-3 is like the other configuration frames and contains much of the same data but with added information and flexible framing. The frame contents are shown in [Table A.9](#) and [Table A.10](#). CFG-3 is optional; a synchrophasor device does not have to implement this message to be considered compliant with the version 2 standard.

Table A.9—Configuration frame 3 organization, for version 2

No	Field	Size (bytes)	Short description
1	SYNC	2	Sync byte followed by frame type and version number.
2	FRAMESIZE	2	Number of bytes in frame, defined in A.2 .
3	IDCODE	2	PMU/PDC data stream ID number, 16-bit integer, defined in A.2 .
4	SOC	4	SOC time stamp, defined in A.2 .
5	FRACSEC	4	Fraction of second and message time quality, defined in A.2 .
6	CONT_IDX	2	Continuation index for fragmented frames.
7	TIME_BASE	4	Resolution of FRACSEC time stamp.
8	NUM_PMU	2	The number of PMUs included in the data frame.
9	STN	1 to 256	Station name—In ASCII format with field index (see Table A.11).
10	IDCODE	2	Data source ID number identifies source of each data block.
11	G_PMU_ID	16	Global PMU ID.
12	FORMAT	2	Data format within the data frame.
13	PHNMR	2	Number of phasors—2-B integer (0 to 32 767).
14	ANNMR	2	Number of analog values—2-B integer.
15	DGNMR	2	Number of digital status words—2-B integer.

Table continues

Table A.9—Configuration frame 3 organization, for version 2 (continued)

No	Field	Size (bytes)	Short description
16	CHNAM	1 to 256 per name	Phasor and channel names—In ASCII with field index (see Table A.11). Minimum of 1 B for each phasor, analog, and digital channel. Names are in the same order as they are transmitted: all phasors, all analogs, and all digitals. For digital channels, the channel name order will be from the least significant to the most significant. (The first name is for bit 0 of the first 16-bit status word, the second is for bit 1, etc., up to bit 15. If there is more than 1 digital status, the next name will apply to bit 0 of the second word and so on).
17	PHSCALE	12 × PHNMR	Conversion factor for phasor channels with flags.
18	ANSCALE	8 × ANNMR	Conversion factor for analog channels.
19	DIGUNIT	4 × DGNMR	Mask words for digital status words.
20	PMU_LAT	4	PMU latitude in degrees, 32-bit floating point, WGS84 datum.
21	PMU_LON	4	PMU longitude in degrees, 32-bit floating point, WGS84 datum.
22	PMU_ELEV	4	PMU elevation in meters, 32-bit floating point, WGS84 datum.
23	SVC_CLASS	1	Service class, as defined in IEEE Std C37.118.1 [B6], a single ASCII character that is M or P for IEEE Std C37.118.1.
24	WINDOW	4	Phasor measurement window length including all filters and estimation windows in effect. Value is in μs, 4-B integer value (to nearest μs).
25	GRP_DLY	4	Phasor measurement group delay including all filters and estimation windows in effect. Value is in μs, 4-B integer value (to nearest μs).
26	FNOM	2	Nominal line frequency code and flags.
27	CFGCNT	2	Configuration change count.
<i>Repeat 9 to 27</i>			Fields 9 to 27, repeated for as many PMUs as in field 8 (NUM_PMU).
28+	DATA_RATE	2	Rate of data transmissions.
29+	CHK	2	CRC-CCITT.

Table continues

Table A.10—Word definitions unique to configuration frame 3

Field	Size (bytes)	Description
SYNC	2	First byte: AA hex Second byte: 52 hex for configuration 3 Frame is version 2.
CONT_IDX	2	Continuation index for fragmented frames: 0: Only frame in configuration, no further frames 1: First frame in series, more to follow 2 to 65534: Number of each succeeding frame, in order 65535 (hex FFFF): Last frame in series
TIME_BASE	4	Same as CFG-1 and CFG-2.
NUM_PMU	2	Same as CFG-1 and CFG-2.
STN	1 to 256	Station name—In UTF-8 format, up to 255 B using the name field format (see Table A.11).
IDCODE	2	Same as CFG-1 and CFG-2.
G_PMU_ID	16	This 128-bit global PMU ID shall be a user-assigned value. It shall be stored in the PMU or other sending device so it can be sent with this configuration 3 message. It allows uniquely identifying PMUs in a system that has more than 65 535 PMUs. The coding for the 16 B is left to the user for assignment.
FORMAT	2	Same as CFG-1 and CFG-2.
PHNMR	2	Same as CFG-1 and CFG-2.
ANNMR	2	Same as CFG-1 and CFG-2.
DGNMR	2	Same as CFG-1 and CFG-2.
CHNAM	1 to 256	Channel names—One name for each phasor, analog, and digital channel in UTF-8 using the name field format (see Table A.11). Names appear in the same order as they are transmitted: all phasors followed by all analogs followed by all digitals. For digital channels, the order is the same as described for configurations 1 and 2.
PHSCALE	12	Magnitude and angle scaling for phasors with data flags. This factor has three 4-B long words: the first is bit-mapped flags, the second is a magnitude scale factor, and the third is an angle offset. First 4-B word. First 2 B: 16-bit flag that indicates the type of data modification when data is being modified by a continuous process. When no modification process is being applied, all bits shall be set to 0. (bit 0 is the LSB, bit 15 is the MSB). This flag shall be used in conjunction with the data modification bit (bit 9) in the status word. That bit shall be set in any frame that data has been modified by any process including those indicated in this flag.

Table continues

Table A.10—Word definitions unique to configuration frame 3 (continued)

Field	Size (bytes)	Description
		Bit # Meaning when bit set 0 Not used, reserved 1 Up sampled with interpolation 2 Upsampled with extrapolation 3 Down sampled by reselection (selecting every Nth sample) 4 Down sampled with FIR filter 5 Down sampled with non-FIR filter 6 Filtered without changing sampling 7 Phasor magnitude adjusted for calibration 8 Phasor phase adjusted for calibration 9 Phasor phase adjusted for rotation ($\pm 30^\circ$, $\pm 120^\circ$, etc.) 10 Pseudo-phasor value (combined from other phasors) 11 to 14 Reserved for future assignment 15 Modification applied, type not here defined Third byte: Phasor type indication (bit 0 is the LSB, bit 7 is the MSB) Bits 07 to 04: Reserved for future use Bit 03: 0 = voltage 1 = current Bits 02 to 00: Phasor component, coded as follows: 111: Reserved 110: Phase C 101: Phase B 100: Phase A 011: Reserved 010: Negative sequence 001: Positive sequence 000: Zero sequence Fourth byte: available for user designation Second and third 4-B words. For phasor $X = X_m e^{j\theta}$, this defines the scaling Y and angle offset θ to be applied to the phasor as follows: $X' = Y \times X_m e^{j(\theta - \theta_0)}$ The second 4-B word is the scale factor Y in 32-bit IEEE floating point. This scales phasor data to primary volts or amperes. If phasors are transmitted in floating-point format and scaled already, this value shall be set to 1. The third 4-B word is the phasor angle adjustment θ in radians represented in 32-bit IEEE floating point. If phasors are transmitted in floating-point format and scaled already, this value shall be set to 0.
ANSCALE	8	Linear scaling for analog channels. For analog value X, this defines scale M and offset B for $X' = M \times X + B$. First 4 B: Magnitude scaling M in 32-bit floating point. Last 4 B: Offset B in 32-bit floating point.
DIGUNIT	4	Mask words for digital status words. Two 16-bit words are provided for each digital word. The first will be used to indicate the normal status of the digital inputs by returning a 0 when exclusive ored (XOR) with the status word. The second will indicate the current valid inputs to the PMU by having a bit set in the binary position corresponding to the digital input and all other bits set to 0. See NOTE.
PMU_LAT	4	PMU latitude in degrees, range -90.0 to $+90.0$. Positive values are N of equator. WGS 84 datum. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.
PMU_LON	4	PMU longitude in degrees, range -179.99999999 to $+180$. Positive values are E of the prime meridian. WGS 84 datum. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.
PMU_ELEV	4	PMU Elevation in meters, Positive values are above mean sea level. WGS 84 datum. Number in 32-bit IEEE floating-point format. For unspecified locations, infinity shall be used.

Table continues

Table A.10—Word definitions unique to configuration frame 3 (continued)

Field	Size (bytes)	Description
SVC_CLASS	1	Service class, as defined in IEEE Std C37.118.1 [B6], a single ASCII character. In 2011 it is M or P.
WINDOW	4	Phasor measurement window length including all filters and estimation windows in effect. Value is in μs , 4-B signed integer value (to nearest μs). (For information only, any required compensation is already applied to the measurement).
GRP_DLY	4	Phasor measurement group delay including all filters and estimation windows in effect. Value is in μs , 4-B signed integer value (to nearest μs). For information only, any required compensation is already applied to the measurement.
FNOM	2	Nominal line frequency code (16-bit unsigned integer). Bits 15 to 1: Reserved Bit 0: 1—Fundamental frequency = 50 Hz 0—Fundamental frequency = 60 Hz
DATA_RATE	2	Rate of phasor data transmissions—2-B integer word (–32 767 to +32 767) If DATA_RATE > 0, rate is number of frames per second. If DATA_RATE < 0, rate is negative of seconds per frame. (E.g., DATA_RATE = 15 is 15 frames per second; DATA_RATE = –5 is 1 frame per 5 s)
CFGCNT	2	Same as CFG-1 and CFG-2.

NOTE—If digital status words are used for something other than Boolean status indications, the use of masks is left to the user, such as min or max settings.

Station and signal names formats are shown in Table A.11.

Table A.11—Name field description

More names	Name field (single station or signal name)		More names
...	1 B	0 B to 255 B	...
—	Name length— unsigned 8-bit integer, 0 to 255	Name, UTF-8 coding Note that standard 7-bit ASCII characters are the same as UTF-8 and no conversions are required	—

A.5 Header frame

This frame shall be human-readable information about the PMU, the data sources, scaling, algorithms, filtering, or other related information. It is shown in Table A.12.

Table A.12—Header frame organization, for version 2

No	Field	Size (bytes)	Comment
1	SYNC	2	Sync byte followed by frame type and version number (AA11 hex).
2	FRAMESIZE	2	Number of bytes in frame, defined in A.2.
3	IDCODE	2	PMU/PDC data stream ID number, 16-bit integer, defined in A.2.
4	SOC	4	SOC time stamp, defined in A.2.
5	FRACSEC	4	Fraction of second and time quality, defined in A.2.
6	DATA 1	1	ASCII character, 1st byte.
K + 6	DATA k	1	ASCII character, Kth byte, K > 0 is an integer.
K + 7	CHK	2	CRC-CCITT.

A.6 Command frame

A data sending device (PMU or PDC) shall be able to receive commands and take appropriate actions. These are shown in Table A.13 and Table A.14.

Table A.13—Command frame organization, for version 2

No	Field	Size (bytes)	Comment
1	SYNC	2	Sync byte followed by frame type and version number (AA41 hex).
2	FRAMESIZE	2	Number of bytes in frame, defined in A.2.
3	IDCODE	2	PMU/PDC ID data stream number, 16-bit integer, defined in A.2.
4	SOC	4	SOC time stamp, defined in A.2.
5	FRACSEC	4	Fraction of second and time quality, defined in A.2.
6	CMD	2	Command being sent to the PMU/PDC (0).
7	EXTFRAME	0 to 65518	Extended frame data, 16-bit words, 0 to 65518 B as indicated by frame size, data user defined.
8	CHK	2	CRC-CCITT.

Table A.14—Commands sent to the PMU/PDC, for version 2

Command word bits	Definition
Bits 15 to 0:	—
0000 0000 0000 0001	Turn off transmission of data frames.
0000 0000 0000 0010	Turn on transmission of data frames.
0000 0000 0000 0011	Send HDR frame.
0000 0000 0000 0100	Send CFG-1 frame.
0000 0000 0000 0101	Send CFG-2 frame.
0000 0000 0000 0110	Send CFG-3 frame (optional command).
0000 0000 0000 1000	Extended frame.
0000 0000 xxxx xxxx	All undesiguated codes reserved.
0000 yyyy xxxx xxxx	All codes where yyyy ≠ 0 available for user designation.
zzzz xxxx xxxx xxxx	All codes where zzzz ≠ 0 reserved.

Annex B

(informative)

Synchrophasor measurements

B.1 Introduction

Synchrophasor measurement definitions, detailed descriptions, and performance requirements are presented in IEC/IEEE 60255-118-1 which was published in December 2018.²² These requirements were first described in IEEE Std 1344 [B4], which also included requirements for transmission of the measurements using real-time communications. The next IEEE Standard for synchrophasors was IEEE Std C37.118-2005 [B5] which also included both measurement and communication requirements. Following that the next revision split requirements into two series: IEEE Std C37.118.1-2011 [B6] includes only measurement requirements and IEEE Std C37.118.2-2011 [B7] includes only communication requirements. This revision of IEEE Std C37.118.2 thus covers only communication aspects.

The purpose of this annex is to introduce the basic terms and concepts of the synchrophasor measurement and the data it produces. The formulas and descriptions provided here are the same as in the IEC/IEEE Std 60255-118-1 standard but abbreviated and simplified for clarity. Consult the standard for details.

B.2 Synchrophasor definition

The voltage or current in an AC power system is nominally a sinusoid which is represented as shown in Equation (1):

$$x(t) = X_m(t)\cos[\theta(t)] + D(t) \quad (1)$$

where

- t is time in seconds, where $t = 0$ is coincident with a UTC second rollover
- X_m is the peak magnitude of the sinusoidal AC signal
- θ is the angular position of the sinusoidal AC signal in radians
- D is a disturbance signal that contains additive contributions to the signal, including, but not limited to harmonics, noise, DC offset and out-of-band interference

Using this definition for the power signals, we define the synchrophasor as the complex number:

$$X(t) = \left(\frac{X_m(t)}{\sqrt{2}}, \phi(t) \right) \quad (2)$$

Where the phase angle $\phi(t)$ is:

$$\phi(t) = \theta(t) - 2\pi f_0 t \quad (3)$$

²²Information on references can be found in [Clause 2](#).

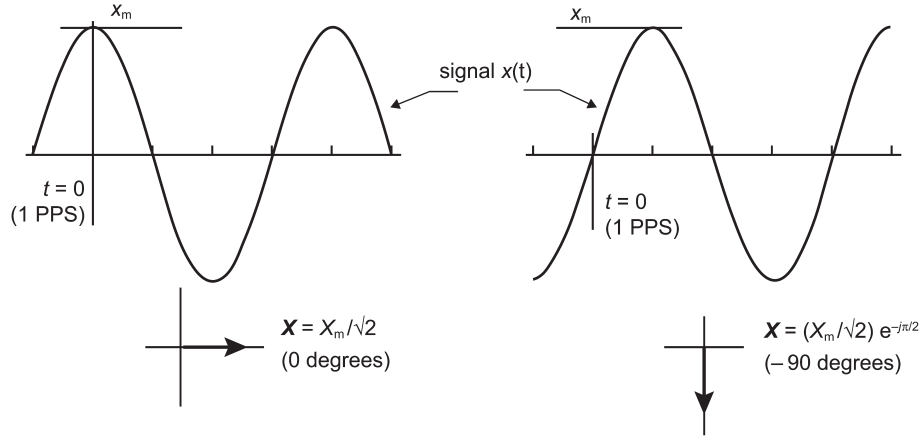


Figure B.1—Convention for synchrophasor representation

B.3 Frequency and rate-of-change-of-frequency definition

Given the definitions above of the power signal and synchrophasor, the frequency and rate-of-change-of-frequency (ROCOF) are given in Equation (4) and Equation (5):

$$f(t) = \frac{1}{2\pi} \frac{d\theta(t)}{dt} = f_0 + \frac{1}{2\pi} \frac{d[\phi(t)]}{dt} \quad (4)$$

$$\text{ROCOF}(t) = \frac{df(t)}{dt} = \frac{1}{2\pi} \frac{d^2\theta(t)}{dt^2} = \frac{1}{2\pi} \frac{d^2\phi(t)}{dt^2} \quad (5)$$

These formulas describe the synchrophasor, frequency, and ROCOF for the given power signal (which could be voltage or current) as defined by the synchrophasor standard. See the Standard IEC/IEEE Std 60255-118-1 for further details.

B.4 Measurement time for synchrophasors

Synchrophasor measurements are tagged with the UTC time corresponding to the time of measurement. UTC has an occasional leap second added to or deleted from the time scale to keep it synchronized with celestial time. In the UTC time scale, insertion of a leap second results in the seconds count going to 60 before roll-over to account for the extra second. When using a second of century (SOC) integer count for seconds, this results in two successive seconds having the same SOC count. This needs to be noted by an extra indicator so measurement data for the extra second can be accounted. Conversely, the deletion of a leap second results in a second that is missing, so the second count only goes to 58 and an indicator is required to maintain data synchronization.

B.5 Synchrophasor transmission²³

Synchrophasor data historically has been sent at a constant rate called a ‘reporting rate’. The rates are set to deliver an integer number of measurements each second, evenly spaced through the second. All measurements are made and adjusted so that the value represents the signal at a specific time. (Note that while we usually just think of making a measurement at a certain time, synchrophasors are estimated from multiple samples

²³Specific requirements for communications are included in Clause 5 of this standard.

made over an interval of time, so the “time of measurement” should be carefully adjusted to best fit the signal). Several measurements may be made at a specific time, such as phasors, frequency, ROCOF, and status (Boolean) values. It is most efficient to send a group of measurements made at the same time together with a common time stamp showing the time of measurement. This measurement group is called a “data frame”; it includes the time stamp of the measurements, the measurements, some framing information, and some quality indicators.

The synchrophasor standard IEC/IEEE Std 60255-118-1 requires making measurements at certain reporting rates (F_s) with the assumption they will be transmitted at one of these rates. The algorithms for making the synchrophasor estimates vary somewhat based on the reporting rate and are specified accordingly. The reporting rates specified by IEC/IEEE Std 60255-118-1 are given in [Table 18](#).

B.6 Synchrophasor measurements

Requirements for synchrophasor, frequency, and ROCOF estimates are detailed in IEC/IEEE Std 60255-118-1. Refer to that standard for measurement requirements.

Annex C

(informative)

Synchrophasor measurement systems

C.1 Synchrophasor network

A simple structure of a synchrophasor network consists of the PMUs and PDCs as shown in Figure C.1. If multiple devices provide synchrophasor measurements in a substation, a local PDC may be deployed there. Typically, multiple PMUs located at key substations gather data and send it live to the local PDC, which in turn sends it to the utility location. The data collected by PDCs may be used to support many applications, ranging from visualization of information and alarms for situational awareness, to ones that provide sophisticated analytical, control, or protection functionality. Dynamic system monitoring applications use full-resolution real-time data along with grid models to support both operating and planning functions. The application displays locally measured frequency, primary voltages, currents, real and reactive power flows, and other quantities for system operators. Many PDCs belonging to different utilities can be connected to a common central PDC to aggregate data across the utilities, to provide an interconnection-wide snapshot of the power grid measurements. Typically, PMUs and local PDCs are in a substation, relatively close to each other. The substation itself may be at a remote location. Physical access to a substation device may cause some inconvenience. This factor has received some attention in [B11], and in this revision of this standard.

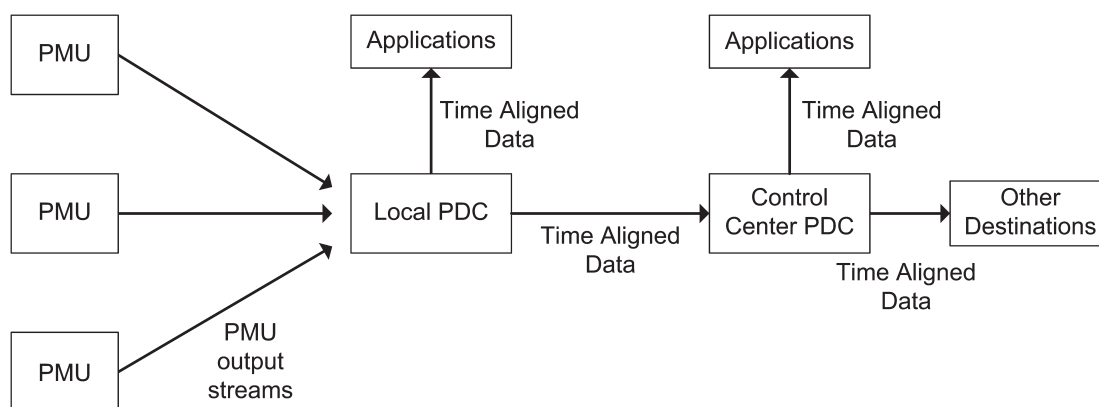


Figure C.1—Synchrophasor data collection network

C.2 Synchrophasor network elements

C.2.1 Phasor measurement unit

The PMU is a function or device that provides synchrophasors and system frequency estimates, as well as other optional information such as calculated megawatts (MW) and megavars (MVAR), and Boolean status words. It provides synchrophasor estimates from one or more voltage or current waveforms. It can be realized as a stand-alone physical device or as a part of a multifunction device such as a protective relay, DFR, or meter. This information may be recorded locally or transmitted in real time to a central location as illustrated in Figure C.1. This standard addresses the real-time transfer of data from the PMU to the PDC or other devices.

C.2.2 Phasor data concentrator

A PDC may be considered a node in a communication network and is defined as a set of functions that combines synchronized phasor data from multiple sources for further processing. The functions may reside in a standalone device or may be software functions on a server/computer. These functions can be found in the standard IEEE Std C37.247™-2019 and the guide IEEE Std C37.244™-2013.

A PDC correlates and combines synchrophasor data from several PMUs or PDCs and feeds out one or more data streams to the higher level PDCs and/or applications. The PDC correlates synchrophasor data by time tag to create a system wide measurement set. It also serves to monitor the overall measurement system health.

A structured hierarchy of distributed PDCs may be formed. Substation PDCs as shown in [Figure C.1](#) aggregate and time-align synchrophasor data from multiple PMUs and feed the data to the control center. Control center PDCs collect synchrophasor data from multiple substation PDCs to establish a wide area measurement system. All PDCs may support application functions local to the PDC, data archiving, etc.

C.2.3 Application functions

PMUs provide system wide synchronized measurements of voltage and current phasors, including absolute angles. The latter ability allows detection of angle differences between distant locations. These features offer many possibilities for determining the state of the system in real-time, and thereby support multiple protection and control applications. Examples of such applications are accurate state estimation, supervision of distance relays, adaptive out of step protection, voltage instability detection, oscillation detection, sequence of event detection, fault location, disturbance recorders, accurate post-mortem analysis of events, power quality monitoring etc. These applications can be used for both transmission and distribution systems.

C.3 Communications

C.3.1 Data streams

A PMU or PDC may transmit its data in one or more separate data streams. Each stream may have different contents, may be sent at a different rate, and may have a different destination. Each stream should be individually controllable and have its own STREAM_ID. This feature is useful for sending data to different destinations/applications with different purposes, allowing latency control and class of service (M and P class) supply.

C.3.2 Protocols

Two active standards exist at present for the communication of PMU data—IEEE Std C37.118.2-2011 [\[B7\]](#) and IEC TR 61850-90-5 [\[B2\]](#).²⁴ The protocol of IEEE Std C37.118-2005 [\[B5\]](#) is widely implemented. Other legacy protocols exist in the field. These include some custom protocols (e.g., BPA/PDCstream) and IEEE Std 1344-1995 [\[B4\]](#). IEEE Std C37.118.2-2011 [\[B7\]](#) was designed to be backward compatible with the 2005 version.

C.3.3 Cyber security

Synchrophasor data networks may include a variety of protocols and communication networks. The latter may include public data channels. This can create vulnerabilities that should be addressed at system level by system designers.

²⁴IEC TR 61850-90-5 [\[B2\]](#) is a technical report showing the use of IEC 61850 to transmit synchrophasor information per IEEE Std C37.118.

Devices comprising synchrophasor data networks should implement functions as dictated by system level requirements for cybersecurity. Users should know what these are and should specify them to the manufacturer. Such specifications of security are beyond the scope of this standard. See IEEE Std C37.240™ [\[B10\]](#) for industry best practices that may be applied to a synchrophasor data network.

Annex D

(informative)

System communications considerations

D.1 Communication bandwidth

This standard does not impose any restriction on the communication system or media itself. It defines three message types—*data*, *configuration*, and *command*—to and from a PMU or PDC for real-time synchrophasor communication. It also prescribes the message contents and format. These are detailed in [Clause 4](#).

Typically, the receiving device requests a configuration using the commands at startup or a configuration change only. Most of the time the data frame is transmitted continuously from the PMU or PDC at the designated reporting rate. Consequently, the required bandwidth is determined by the data frame size, data rate, and communication overheads. The frame size is precisely described in this standard and varies depending on the number of phasors and analog and digital words included in the frame, as seen in [Table 4](#). Typical sizes vary from 40 B to 70 B for frames from a single PMU to over 1000 B in a frame from a PDC with data from many PMUs. [Table D.1](#) gives a formula to calculate the transmission rate in bytes per second (BPS), and examples for some typical PMU configurations. Note that the communication media will add overhead, which can be considerable. Asynchronous serial communication adds 2 bits per byte (25% overhead) and TCP and UDP add 54 B and 42 B/frame respectively, which can amount to more than 50% overhead.

Table D.1—Estimation of bandwidth requirements for example PMU configurations

Quantity	Description
Transmission rate (bytes/second)	$\text{DATA_RATE (PMU or STREAM)} \times (\text{TCP/UDP_OH} + 118_OH + 118_PAYLOAD)$
DATA_RATE (PMU or STREAM)	Defined in Table 3 . Number of data frames transmitted each second.
TCP/UDP_OH	TCP = 54 B, UDP = 42 B. This is the fixed number of bytes added to each data frame by IP protocols.
118_OH	16 B (see Figure 2 and Table 1). This is the fixed number of bytes in each data frame, apart from the measured values, and includes SYNC, FRAMESIZE, STREAM_ID (or PMU_ID), SOC, LEAP_BYTE, FRACSEC, and CHK word.
118_PAYLOAD	(STAT_FLAG WORD = 2 B) + (TIME_QUALITY = 2 B) + (PHNMR $\times$ bytes/phasor) + (ANNMR $\times$ bytes/analog) + (FRNMR $\times$ bytes/freq) + (DFDTNMR $\times$ bytes/dfdt) + (DGNMR $\times$ 2)
PHNMR, ANNMNR, FRNMR, DFDTNMR, DGNMR	These are defined in Table 3 and indicate the number of specified measurements included in the data frame.
Bytes/quantity (various)	These are determined by the data format used for the specific measurement included in the data frame. FORMAT field in Table 3 defines these.

D.2 Communication delay

An interval of time is required to make a measurement, send the measurement data to a device that will use the measurement, and then use the data. This interval of time from when the signal has a certain value to when the measurement is consumed by the application is called *delay* in this subclause. The first aspect of this delay is in the measurement process. That includes the window over which data is gathered to make a measurement,

the estimation method and filtering, and the time to process that data. The second aspect is the communication of the data including the time to send each bit of the message, the distance and type of communication, and various communication buffering, multiplexing, conversion processes. Finally, the last aspect is the processing of the receiving device and application algorithms.

Delay in measurement is largely dependent on the processing window and filtering, which vary with the data reporting rate and the PMU class of service. Processing delays for calculating the measurement are generally very small compared with other delays. Additional delays caused by servicing other processing needs such as a multifunction device may cause large interruptions or may be insignificant.

Communication delay starts with clocking the bits out of the PMU and is easily measured. With RS-232 serial this delay can be in the millisecond range, but at network speeds is insignificant. Once multiplexed onto a communication baseband, data travels at 0.6 to 0.99 times the speed of light. For example, data sent 500 km would only be delayed 2 ms to 3 ms, a relatively small number that is predictable based on the communication network. However, buffering, de-multiplex/multiplex points, forwarding, and routing can add tens to hundreds of milliseconds and may not be very predictable. Error detection/correction can add hundreds of milliseconds to seconds. Data concentration with delays allowed for data with various delays can also add seconds. Some typical values for delays are summarized in [Table D.2](#).

Table D.2—Summary of delays' causes and typical ranges

Cause of delay	Typical range of delay
Sampling window (delay 1 / 2 of window)	17 ms to 100 ms
Measurement filtering	8 ms to 100 ms
PMU processing	0.005 ms to 30 ms
PDC processing and alignment	2 ms to 2+ s
Serializing output	0.05 ms to 20 ms
Communication system I/O	0.05 ms to 30 ms
Communication distance	3.4 μ s/km to 6 μ s/km
Communication system buffering and error correction	0.05 ms to 8 s
Application input	0.05 ms to 5 ms

Note that while these are typical, actual values can be different and systems need to be evaluated on a case-by-case basis. The minimum overall delay one could anticipate is in the order of 20 ms (no added filtering). A typical system from PMU to PDC will be in the 20 ms to 50 ms range. Each level above that will add some PDC processing and wait times, probably in the 30 ms to 80 ms range (and in addition to the PMU to PDC time). The typical delay in a real-time system is in the 0.2 s to 2 s range but the maximum could be much longer. If delay in data delivery is critical to applications, the designers need to examine all communication aspects they are incorporating to make sure delivery time meets specifications and there are no unexpected delays.

D.3 Synchrophasor message mapping into communications

As communications have evolved into network methods, synchrophasor communication has moved as well. Several common methods using IP have evolved. These methods have been followed by vendors and implementers. IP is widely used and commonly available, so these methods are easy to implement with standard equipment and easy to operate by IT personnel. These methods are described in the following subclauses.

D.3.1 Network communications using IP

Phasor messages shall be mapped in their entirety into TCP (as defined in RFC 793 [B14]) or UDP (as defined in RFC 768 [B12]). They shall be written to and read from using standard IP input-output functions. Default port numbers shall be 4712 for TCP and 4713 for UDP, but in all cases, the user shall be provided the means to set port numbers as desired (see Table 3). The IP may be carried over Ethernet or another transport means. With a stacked protocol like IP, each message layer is encapsulated in the next one down to the link and physical layer where the message is sent. The physical layer is to convert the digital signals to the required physical medium signals (e.g., electrical, fiber, wireless). This process is illustrated in Figure D.1.

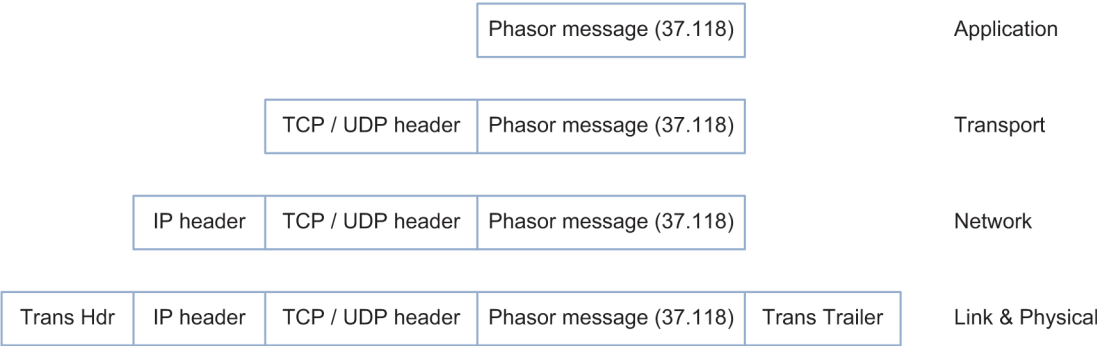


Figure D.1—Mapping of IEEE C37.118 data into a TCP or UDP packet—A transport layer header and trailer are shown, as it would be when using Ethernet

D.3.2 Synchrophasor communication methods for IP

D.3.2.1 Transmission using IP over Ethernet

Phasor measurement systems most commonly use the IP over network communications. The previous standard, IEEE Std C37.118-2005 [B5], as well as this standard require that if a stacked protocol such as IP is used for transmitting the messages, they are included in their entirety (all parts of the defined message are included). It further prescribes default ports for using communication by both TCP and UDP, though the user can specify any port assignment they prefer.

The common methods for communication are as follows:

- a) Client-server: The device providing data is the server and the device receiving data is the client. The device providing data can be a PMU, a PDC, or any other device that will output synchrophasor data. The device receiving data can be a PDC or any or other device that receives synchrophasor data. In cases where data transmission is initiated by command, the client initiates contact and controls data flow with commands. With commanded operation, the server only sends data when a client requests it using the standard Start and Stop commands. Both modes may support commands to retrieve configuration data. The client-server mode can transport synchrophasor data by either TCP or UDP, or a combination of both.
- b) Spontaneous: With spontaneous, the server sends data by UDP to a designated destination without stopping, whether a receiving device is present or not. The stream is initiated by a function in the device accessed separately from data operations.
- c) Multicast: This method sends data to multiple recipients in a single transmission [B15].

D.3.2.2 TCP-only method

This method has a single TCP connection over which commands, data, and configuration frames are passed. The client needs to know only the server address and port. The usual TCP data management advantages and benefits apply to all data. The link is easy to administer, troubleshoot, and manage. The main disadvantage is that with the high rate and continuous data transmission, a single dropped data packet can cause data backup in the TCP mechanism that causes an interval of data loss. If the data is recovered, all data will be delayed, which is often much worse for real-time applications than the loss of one packet. There is also more traffic generated by the return acknowledgments. TCP is a connection-oriented protocol, thus is strictly a 1-to-1 connection, so if data is required for more than one client, it should be sent separately to each destination. This increases traffic and requires the PMU to support multiple TCP connections. Another TCP disadvantage comes from the fact it requires two-way communication to make a connection; if security measures block incoming traffic from a protected zone, a TCP client will be unable to connect.

D.3.2.3 UDP-only method

This method uses straight UDP for communication in both directions for PMU messages, including commands, data, and configuration. The client should know the server address and port number. The server can respond to the client port or a different port by prior arrangement. The advantages of using UDP are reduced bandwidth requirements and elimination of a backup delay, due to dropouts as described for TCP in [D.3.2.2](#). The disadvantage is that server-client communications are not confirmed so it is difficult to locate problems if the communications do not proceed as expected. With UDP, data is not retransmitted in case of error, so errored packets are permanently lost (unless a suitable recovery system is provided by the user). There also can be difficulty with controlling data streams since more than one client can access the connectionless control port. If data is sent to a unicast IP address, it then should be sent to each client separately. Alternatively, data can be sent to a multicast IP address, allowing numerous clients to receive it, thus minimizing network traffic. However, the user will need to set the multicast address and port separately since the command does not do this.

D.3.2.4 TCP/UDP method

This method uses TCP for commands, and configuration communications, and UDP for sending data. The server address and port should be known to the client. If the UDP is sent by multicast, that address should be known to the server as well. The commands, and configuration communication are transmitted by TCP and the data is sent by UDP. This reduces data transmission delay and does not risk backup, though data will not be replaced if corrupted. This method has the advantage of more secure transmission on the critical data portions, and minimal delay or backup risk on the streaming data. It can also use multicast to reduce traffic and can support retrieving configuration information without interrupting the data transmission.

D.3.2.5 Spontaneous data transmission method

With this method, a PMU, PDC, or other data serving device sends out data in the IEEE C37.118.2 format to a preset destination continuously. The data is sent by UDP and can be unicast or, multicast. The output is initiated by a device setting or device interface—not through the IEEE C37.118.2 command set. To be compliant with the standard, command functionalities for retrieving the PMU configuration should be provided. While this will make the device technically non-compliant, the user may prefer the benefits of this operation. Since the data is sent by UDP, a destination device does not have to be present. This mode is useful for sending data using multicast to many clients that may be coming online and offline randomly. It allows a steady supply of data without interruption and can be used where two-way communication is prohibited, such as through a firewall. Header and configuration information retrieval may be allowed on the same or a different channel. In some cases, a stream configuration frame is sent spontaneously at a preset interval, such as once a minute, along with the data. The drawback to this method is lack of ability to turn the data stream on and off. Since the PMU is preconfigured with the client Ips or a multicast IP, changing a client requires reconfiguring the PMU devices.

Annex E

(informative)

Configuration example

E.1 A simple configuration method

This revision of the standard adds the ability to configure devices in a hierarchical synchrophasor data network, such as the one shown in [Figure 1](#). The added mechanism (capability, rename, and configure stream frames) allows an operation like the publish/subscribe mechanism. Without this ability, each device should be configured independently. If the devices came from different manufacturers, it also means different configuration software tools to be used for different devices. There is a significant potential for inadvertent errors.

This annex describes a simple step by step method to set up a conventional synchrophasor data network, as shown in [Figure E.1](#) and [Table E.1](#). Basic communications should be established between devices before any method can be used. Users may devise other methods that may include PMUs communicating directly with the control center PDC.

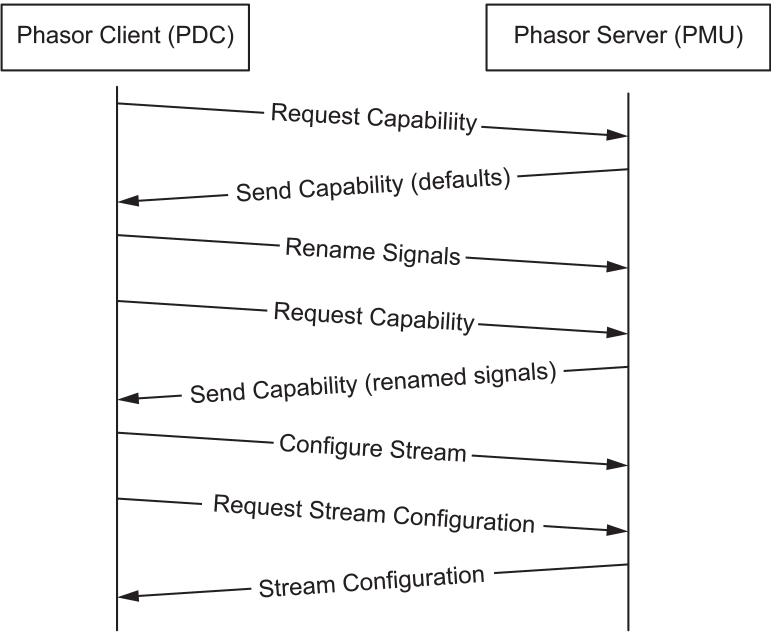


Figure E.1—Configuration command structure

Table E.1—A simple step by step configuration method

Assigning signal names	
1	PDC requests capability frame from the PMU.
2	PMU sends capability frame to the PDC. These will all be default names, as shipped from the factory.
3	PDC sends rename signals frame to PMU.
4	PDC requests capability frame from the PMU.
5	PMU sends capability frame to the PDC. These will include renamed signals. This is confirmation for the PDC that the rename signal frame has been acted upon.
Data stream configuration	
6	PDC sends configure stream frame to PMU.
7	PDC requests stream configuration from the PMU.
8	PMU sends stream configuration to the PDC. This is confirmation for the PDC that the configure stream frame has been acted upon.

E.2 Command splitting at the substation PDC

If a valid command from a control center requires that a change is needed in a PMU signal (such as a rename command), the substation PDC may split the control center command (a configuration frame) into a corresponding rename command to a PMU. The most significant use of this feature is expected to be during the initial installation, when all PMUs are likely to power up with default signal names.

E.3 PMUs with remote configuration disabled

Some users may choose to disable remote configuration for their PMU devices for security reasons. Such devices may be configured locally and need not be a part of this method. This standard includes a configuration flag for PMUs to indicate this to PDCs. PDCs/control centers can exclude such devices from this method.

Users may choose to not implement this method between a substation PDC and PMUs in that substation. PMUs may be configured to send all their signals to the substation PDC, and the devising of multiple data streams may be implemented only between the substation PDC and the control center.

Annex F

(informative)

Leap second bit timing examples

A leap second is a 1 s adjustment applied occasionally to UTC. This allows the UTC to stay close to the mean solar time. Due to irregularities in the earth's rotation, leap seconds' occurrence is also irregular. These adjustments are generally decided six months in advance by the International Earth Rotation and Reference Systems Service (IERS).

The following examples show how the time count and leap second bits will appear for both a positive (added) leap second and a negative (deleted) leap second. The direction bit can be at any state before the leap second pending bit is set and after the leap second occurred bit clears. When either of these bits is set, the direction bit shall be set in the state properly indicating insertion or deletion. The pending bit shall be set as soon as a leap second occurrence is known but not more than 61 s or less than 1 s before the change will occur, as shown in the example below.

Added leap second:

SOC time	Time of day	Direction bit 6	Occurred bit 5	Pending bit 4	Comments
1114991939	23:58:59	X	0	0	Direction bit any state before pending.
1114991940	23:59:00	0	0	1	Pending can be set no earlier than here.
1114991941	23:59:01	0	0	1	
.....					
1114991998	23:59:58	0	0	1	
1114991999	23:59:59	0	0	1	Pending and direction shall be set by here.
1114992000	23:59:60	0	0	1	Leap second occurs here.
1114992000	00:00:00	0	1	0	Occurred and direction remain set.
1114992001	00:00:01	0	1	0	
.....					
1115078399	23:59:59	0	1	0	
1115078400	00:00:00	X	0	0	Occurred cleared, direction don't care.
.....					

Deleted leap second:

SOC Time	Time of day	Direction bit 6	Occurred bit 5	Pending bit 4	Comments
1114991939	23:58:59	X	0	0	Direction bit any state before pending.
1114991940	23:59:00	1	0	1	Pending can be set no earlier than here.
1114991941	23:59:01	1	0	1	
.....					
1114991997	23:59:57	1	0	1	
1114991998	23:59:58	1	0	1	Pending and direction shall be set by here.
1114992000	00:00:00	1	1	0	Leap second occurred, pending cleared.
1114992001	00:00:01	1	1	0	Occurred and direction remain set.
1114992002	00:00:02	1	1	0	
.....					
1115078399	23:59:59	1	1	0	
1115078400	00:00:00	X	0	0	Occurred cleared, direction don't care.
.....					

Annex G

(informative)

FRACSEC/TIME_BASE usage examples

G.1 Periodic data frames

This standard requires PMUs to calculate the time stamp as a second of century (SOC) and a fractional second value (FRACSEC), as shown in 4.3.1.4. FRACSEC is a 24-bit unsigned integer, calculated on the base of TIME_BASE, and as close to the intended fractional second as possible. For a periodic data frame, the calculated time may not be a rational number, depending on the data reporting rate. Thus, the FRACSEC included in the frame is rounded to the nearest integer. This is illustrated in Table G.1, for a data reporting rate of 30 frames per second for a 60 Hz system.

Table G.1—Rounding-off of FRACSEC values (30 FPS / 60 Hz)

Frame sequence number	Actual fractional seconds	FRACSEC (calculated values)	
		TIME_BASE = 720	TIME_BASE = 1000
0	0.0	0 (0.0)	0 (0.0)
1	0.033..	24 (24.0)	33 (33.3333..)
2	0.066..	48 (48.0)	67 (66.6666..)
3	0.1	72 (72.0)	100 (100.0)
...	—	—	—
28	0.933..	672 (672.0)	933 (933.3333..)
29	0.966..	696 (696.0)	967 (966.6666..)

Selecting TIME_BASE to be an integer multiple of the data reporting rate can help ensure that calculated values for fractional seconds for periodic data frames are always integers.

A similar case can be made for a 50 Hz system and a data reporting rate of 25 frames per second. In this case, a TIME_BASE value of 1000 will yield integer numbers, while a value of 720 will need rounding, as shown in Table G.2.

Table G.2—Rounding-off of FRACSEC values (25 FPS / 50 Hz)

Frame sequence number	Actual fractional seconds	FRACSEC (calculated values)	
		TIME_BASE = 720	TIME_BASE = 1000
0	0.0	0 (0.0)	0 (0.0)
1	0.04	29 (28.8)	40 (40.0)
2	0.08	58 (57.6)	80 (80.0)
3	0.12	86 (86.4)	120 (120.0)
...	—	—	—
23	0.92	662 (662.4)	920 (920.0)
24	0.96	691 (691.2)	960 (960.0)

G.2 Discrete event data frames

Discrete event data frames are not restricted to being periodic. FRACSEC values for successive events can be as close as being consecutive integers. The time resolution between these is the reciprocal of TIME_BASE. A higher TIME_BASE value will allow discrete events to be recorded with a greater time resolution. A few examples are shown in Table G.3 and Table G.4. These show three events 0.1 ms apart, and how different values of TIME_BASE affect the FRACSEC values transmitted, and how the receiver will convert these to event time values. A higher TIME_BASE will allow a better time resolution for recording discrete events.

The values marked with an asterisk (*) cannot be distinguished from others at the receiving end.

Table G.3—Discrete event FRACSEC examples for 50 Hz system

Actual event time	FRACSEC values	
	TIME_BASE = 1000 Resolution = 1 ms	TIME_BASE = 10000 Resolution = 0.1 ms
..	—	—
..	—	—
0.2372	237 *	2372
0.2373	237 *	2373
0.2374	237 *	2374
..	—	—
..	—	—

Table G.4—Discrete event FRACSEC examples for 60 Hz system

Actual event time	FRACSEC values		
	TIME_BASE = 720 Resolution = 1.388.. ms	TIME_BASE = 7200 Resolution = 0.1388.. ms	TIME_BASE = 10020 Resolution = 0.09980 .. ms
..	—	—	—
..	—	—	—
0.2372	171 *	1708	2377
0.2373	171 *	1709 *	2378
0.2374	171 *	1709 *	2379
..	—	—	—
..	—	—	—

Annex H

(informative)

Usage of mixed version PMUs

This version of the standard includes changes that may present compatibility and configuration challenges to systems that include devices using different versions of the C37.118.2 standard. Many systems have been installed and in use with version 2. These will continue to function for several years to come. It will be critical for users/applications to be able to use the two versions together, and to translate from one to the other as far as the differences between them permit. This annex provides guidance for such systems. Systems without a mix of different version devices should run seamlessly and are not addressed here.

A PMU may support either, or both versions of the standard, and use separate data streams for each. It may not be necessary for a PMU to include any conversion or merging functions.

A PDC (substation or control center), or an application may be connected to two or more PMUs using a mix of versions 2 and 3. A PDC should be able to support both versions. A PDC should also be able to combine data from the two PMUs, and to create a data stream in the version required by the destination device/application for that data. It should be able to convert from one version to the other as best as possible, since an exact one to one conversion may not be possible due to the differences in the two versions.

Three configurations are shown. In each case, PMU devices use different versions (version 2 and version 3). The substation PDC and the control center PDC may use different versions of this standard.

H.1 Determining versions

While working with different PMUs with different versions, a PDC will need to determine the versions of the devices it is dealing with. A revision 3 PDC should send a “send capability” (See 4.4.1.1) command to the PMU. If it responds correctly, it is a revision 3 PMU. If it does not respond, the PDC should send a “send cfg1” or a “send cfg2” command (see Annex A). If it responds correctly, it is a revision 2 PMU.

A revision 2 PDC should send a “send cfg1” or a “send cfg2” command. If the PMU responds correctly, the PMU supports revision 2, and should be used accordingly.

In either case, no response could indicate a communications or other problem. In all cases, the PDC should check the PMU version in the SYNC WORD of the received response, as confirmation.

H.2 Substation PDC using version 2

Figure H.1 shows a system with a substation PDC using version 2. In this case, the system can only support version 2 data streams regardless of the version used by the control center PDC. The PMU using version 3 will not be able to communicate with the substation PDC.

H.3 Substation PDC using versions 2 and 3, control center using version 2

Figure H.2 shows a system with the substation PDC using versions 2 and 3, while the control center PDC is using version 2. In this case, both version 2 and version 3 PMUs can communicate with the substation PDC. However, to send these data and configurations to the version 2 control center, the substation PDC extracts

version 2 compliant items from the incoming version 3 stream. The substation PDC also converts version 2 commands from the control center for the version 3 PMU, where supported.

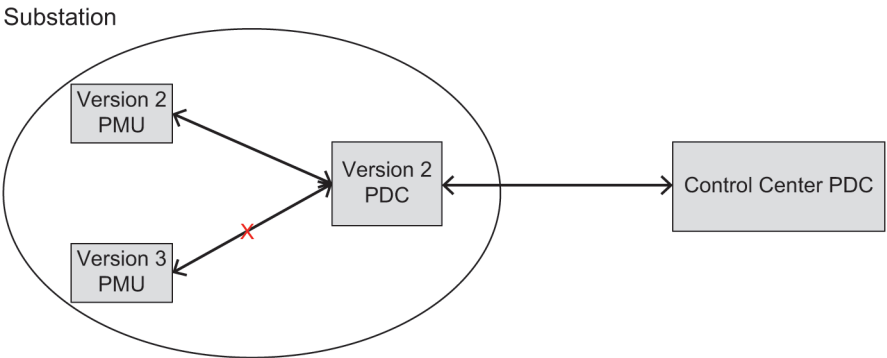


Figure H.1—Substation PDC using version 2, control center PDC version irrelevant

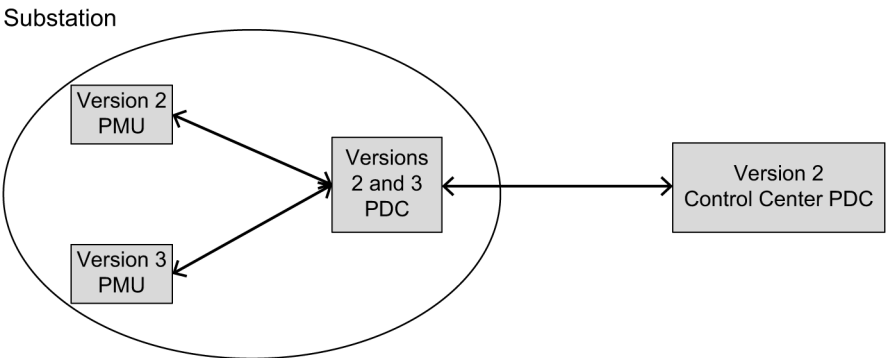


Figure H.2—Control Center PDC using version 2

H.3.1 Commands conversion from version 2 to version 3

A version 2 command can be converted to a version 3 command as shown in [Table H.1](#).

Table H.1—Commands conversion from version 2 to version 3

Version 2 command description	Version 3 command
Turn off transmission of data frames (0x0001).	Turn off transmission of data frames (0x0010).
Turn on transmission of data frames (0x0002).	Turn on transmission of data frames (0x0020).
Send HDR frame (0x0003).	Not supported.
Send CFG-1 frame (0x0004).	Send capability (0x0040).
Send CFG-2 frame (0x0005).	Send stream configuration frame (0x0060).
Send CFG-3 frame (optional command) (0x0006).	Send stream configuration frame (0x0060).
Extended frame (0x0008).	None in version 2.

H.3.2 Config frame conversion from version 3 to version 2

To merge version 3 PMU configuration frames into version 2 configuration frames sent to the control center PDC, the substation PDC may insert/delete corresponding fields as shown in [Table H.2](#).

Table H.2—Configuration frame conversion from version 3 to version 2

Incoming version 3	Outgoing version 2	Comment
Field	Field	
CONT_IDX	CONT_IDX	A calculated value. No change in function.
TIME_BASE	TIME_BASE	No change.
PDC_NAME	—	Delete.
NUM_PMU	NUM_PMU	No change.
—	—	Begin per PMU fields.
PMU_NAME	STN	No change, although the field name has changed.
PMU_ID	IDCODE	No change.
PMU_VERSION	—	Delete.
G_PMU_ID	G_PMU_ID	No change.
FORMAT	FORMAT	No change.
PHNMR	PHNMR	No change.
ANNMR	ANNMR	No change.
DGNMR	DGNMR	No change.
FRNMR	—	Delete. Version 2 includes exactly one measurement.
DFDTNMR	—	Delete. Version 2 includes exactly one measurement.
CHNAM	CHNAM	No change.
PHSCALE 16 × PHNMR	PHSCALE 12 × PHNMR	Delete the fourth 4-B word (voltage class). Version 2 does not support this.
FRSCALE	—	Delete.
DFDTSSCALE	—	Delete.
ANSSCALE	ANSSCALE	No change.
DIGUNIT	DIGUNIT	No change.
PMU_LAT	PMU_LAT	No change.
PMU_LON	PMU_LON	No change.
PMU_ELEV	PMU_ELEV	No change.
PMUFLAG 2 B	SVC_CLASS 1 B	Bits 15 to 4: Set to 0. Bits 3 to 0: Set in accordance with the incoming bit 3 to 0.
WINDOW	WINDOW	No change.
GRP_DLY	GRP_DLY	No change.
PMU_DATA_RATE	—	Set to the same value as the incoming DATA_RATE.
—	FNOM 2 B	To be set by PDC, in accordance with the incoming bit 13 in the PMUFLAG field.
CFGCNT	CFGCNT	No change.
—	—	End per PMU fields.
STREAM_DATA_RATE	DATA_RATE	No change, although the field name has changed.
WAIT_TIME	—	Delete.
CHK	CHK	A calculated value. No change in function.

H.3.3 Periodic data frame conversion from version 3 to version 2

To merge version 3 PMU periodic data frames into version 2 data frames sent to the control center PDC, the substation PDC may delete corresponding fields as shown in [Table H.3](#). Discrete event data frames from version 3 PMUs cannot be included in substation PDC's version 2 output. However, the substation PDC may archive these.

Table H.3—Data frame conversion from version 3 to version 2

Incoming Version 3	Outgoing Version 2	Comment
Field	Field	
STAT_FLAG	STAT	See H.3.4 .
TIMEQUALITY	—	Delete.
PHASOR + (DAPHASOR)	PHASORS	Delete data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.2 .
FREQ + (DAFREQ)	FREQ	Delete data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.2 .
DFREQ + (DADFREQ)	DFREQ	Delete data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.2 .
ANALOG	ANALOG	No change.
DIGITAL	DIGITAL	No change.

H.3.4 STAT_FLAG word conversion from version 3 to version 2

A substation PDC may construct the outgoing version 2 STAT word from the incoming version 3 STAT_FLAG word as shown in [Table H.4](#).

Table H.4—Partial conversion of version 3 STAT_FLAG word to version 2

Outgoing Version 2	Description	Action to take on incoming version 3 STAT_FLAG word
Bits		
15 to 14	Data error	Copy from bits 15 to 14 of incoming STAT_FLAG word.
13	PMU sync	Copy from bit 13 of incoming STAT_FLAG word.
12	Data sorting	Copy from bit 12 of incoming STAT_FLAG word.
11	Trigger	Set to 0. Feature not included in version 3.
10	Configuration change	Set to 0. Feature not included in version 3.
9	Data modified	Set to 0 if any incoming data attribute indicates that data has been modified.
8 to 6	PMU time quality	Calculate from bits 14 to 0 of incoming TIMEQUALITY flag (see 4.5.1.2).
5 to 4	Unlocked time	Set to 0. Not available in version 3.
3 to 0	Trigger reason	Set to 0. Feature not included in version 3.

H.4 Substation PDC using versions 2 and 3, control center using version 3

In this case, the version 2 PMU can communicate with the substation PDC. However, to send these data and configurations to the version 3 control center, the substation PDC extracts version 3 compliant items from the incoming version 2 stream. The items added in version 3 are omitted or set to default, as appropriate. The substation PDC also converts version 3 commands from the control center for the version 2 PMU, where supported.

H.4.1 Commands conversion from version 3 to version 2

Version 2 PMUs do not accept remote configuration commands. If the control center PDC sends such commands, the substation PDC may respond with an error response (see [4.7](#)). These PMUs are set up

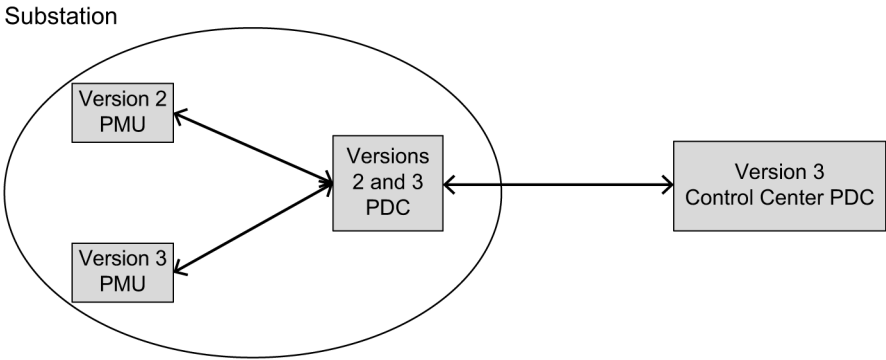


Figure H.3—Control center PDC using version 3

independently, as per current practice. Commands supported by both versions may be converted as shown in Table H.5.

Table H.5—Commands conversion from version 3 to version 2

Version 3 command description	Version 2 command
Turn off transmission of data frames (0x0010).	Turn off transmission of data frames (0x0001).
Turn on transmission of data frames (0x0020).	Turn on transmission of data frames (0x0002).
Send capability (0x0040).	Send CFG-1 (0x0004).
Send stream configuration frame (0x0060).	Send CFG-2 (0x0005), and Send CFG-3 (0x0006). ^a
Extended frame. (0x0080). Old data request	To be omitted. Not supported in version 2.
^a CFG-3 is optional in version 2. There may be no response if the PMU does not support it. Users may use both “Send CFG2” and “Send CFG3” commands in such cases.	

H.4.2 CFG-3 Config frame conversion from version 2 to version 3

To merge version 2 PMU CFG-3 configuration frames into version 3 configuration frames sent to the control center PDC, the substation PDC may insert corresponding fields as shown in Table H.6.

Table H.6—Configuration frame conversion from version 2 to version 3

Incoming version 2	Outgoing version 3	Comment
Field	Field	
CONT_IDX	CONT_IDX	A calculated value. No change in function.
TIME_BASE	TIME_BASE	No change.
	PDC_NAME	Add.
NUM_PMU	NUM_PMU	No change.
—	—	Begin per PMU fields.
STN	PMU_NAME	No change, although the field name has changed.
IDCODE	PMU_ID	No change.
—	PMU_VERSION	Add, from bits 3 to 0 of the incoming SYNC field.
G_PMU_ID	G_PMU_ID	No change.
FORMAT	FORMAT	No change.
PHNMR	PHNMR	No change.

Table continues

Table H.6—Configuration frame conversion from version 2 to version 3 (continued)

Incoming version 2	Outgoing version 3	Comment
Field	Field	
ANNMR	ANNMR	No change.
DGNMR	DGNMR	No change.
—	FRNMR	Set to 1. Version 2 includes exactly one frequency measurement.
—	DFDTNMR	Set to 1. Version 2 includes exactly one ROCOF measurement.
CHNAM	CHNAM	No change.
PHSCALE 12 × PHNMR	PHSCALE 16 × PHNMR	Fourth 4-B word (voltage class) is set to 0.0 since it is not available in Version 2.
—	FRSCALE	Set to 1.0 and 0.0 for scale and offset, respectively.
—	DFDTSIZE	Set to 1.0 and 0.0 for scale and offset, respectively.
ANSCALE	ANSCALE	No change.
DIGUNIT	DIGUNIT	No change.
PMU_LAT	PMU_LAT	No change.
PMU_LON	PMU_LON	No change.
PMU_ELEV	PMU_ELEV	No change.
SVC_CLASS 1 B	PMUFLAG 2 B	Bit 15s to 14: Set to 0. Bit 13: Set in accordance with the incoming FNOM field. Bit 12: Set to 0. Bit 11: To be set by substation PDC, if data is available for retrieval. Bits 10 to 4: Reserved. Set to 0. Bits 3 to 0: Set in accordance with the incoming SVC_CLASS field.
WINDOW	WINDOW	No change.
GRP_DLY	GRP_DLY	No change.
—	PMU_DATA_RATE	Set to the same value as the incoming DATA_RATE.
FNOM 2 B	—	Delete.
CFGCNT	CFGCNT	No change.
—	—	End per PMU fields.
DATA_RATE	STREAM_DATA_RATE	No change, although the field name has changed.
—	WAIT_TIME	Add.
CHK	CHK	A calculated value. No change in function.

H.4.3 CFG-2 Config frame conversion from version 2 to version 3

To merge version 2 PMU CFG-2 configuration frames into version 3 configuration frames sent to the control center PDC, the substation PDC may insert corresponding fields as shown in [Table H.7](#).

Table H.7—Configuration frame conversion from version 2 to version 3

Incoming version 2	Outgoing version 3	Comment
Field	Field	
—	CONT_IDX	A calculated value. No change in function.
TIME_BASE	TIME_BASE	No change.
—	PDC_NAME	Add.
NUM_PMU	NUM_PMU	No change.

Table continues

Table H.7—Configuration frame conversion from version 2 to version 3 (continued)

Incoming version 2	Outgoing version 3	Comment
Field	Field	
—	—	Begin per PMU fields.
STN	PMU_NAME	No change, although the field name has changed.
IDCODE	PMU_ID	No change, although the field name has changed.
—	PMU_VERSION	Add, from bits 3 to 0 of the incoming SYNC field.
—	G_PMU_ID	Set to 0.
FORMAT	FORMAT	No change.
PHNMR	PHNMR	No change.
ANNMR	ANNMR	No change.
DGNMR	DGNMR	No change.
—	FRNMR	Set to 1. Version 2 includes exactly one frequency measurement.
—	DFDTNMR	Set to 1. Version 2 includes exactly one ROCOF measurement.
CHNAM	CHNAM	No change.
PHUNIT	PHSCALE 16 × PHNMR	First 4-B word. First two bytes set to 0. Third byte. Bits 7 to 4: set to 0. Bit 3: 0 = voltage, 1 = current, based on the most significant byte of the received PHUNIT field. Bits 2 to 0: Set to 111. Fourth byte: Available for user designation. Second 4-B word is set to 1.0 if the format for phasor signals is floating point. It is calculated from the least significant three bytes of the received PHUNIT field if the format for phasor signals is fixed point. Third 4-B word is set to 0.0. Fourth 4-B word (voltage class) is set to 0.0 since it is not available in version 2.
—	FRSCALE	Set to 1.0 and 0.0 for scale and offset, respectively.
—	DFDTSKALE	Set to 1.0 and 0.0 for scale and offset, respectively.
ANUNIT	ANSKALE	Substation PDC to add. Set to 1.0 and 0.0 if values are not available.
DIGUNIT	DIGUNIT	No change.
—	PMU_LAT	Set to NaN.
—	PMU_LON	Set to NaN.
—	PMU_ELEV	Set to NaN.
—	PMUFLAG 2 B	Bits 15 to 14: Set to 0. Bit 13: Set in accordance with the incoming FNOM field. Bit 12: Set to 0. Bit 11: To be set by substation PDC, if data is available for retrieval. Bits 10 to 4: Reserved. Set to 0. Bits 3 to 0: Set to 111.
—	WINDOW	Set to −1.
—	GRP_DLY	Set to −1.
—	PMU_DATA_RATE	Set to the same value as the incoming DATA_RATE.
FNOM 2 B	—	Delete.
CFGCNT	CFGCNT	No change.
—	—	End per PMU fields

Table continues

Table H.7—Configuration frame conversion from version 2 to version 3 (continued)

Incoming version 2	Outgoing version 3	Comment
Field	Field	
DATA_RATE	STREAM_ DATA_RATE	No change, although the field name has changed.
—	WAIT_TIME	Add.
CHK	CHK	A calculated value. No change in function.

H.4.4 Data frame conversion from version 2 to version 3

To merge version 2 PMU data frames into version 3 periodic data frames sent to the control center PDC, the substation PDC may insert corresponding fields as shown in [Table H.8](#).

Table H.8—Data frame conversion from version 2 to version 3

Incoming Version 2	Outgoing Version 3	Comment
Field	Field	
STAT	STAT_FLAG	See H.4.5 .
	TIMEQUALITY	See Table 7 .
PHASORS	PHASOR + (DAPHASOR)	Exclude data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.7 .
FREQ	FREQ + (DAFREQ)	Exclude data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.7 .
DFREQ	DFREQ + (DADFREQ)	Exclude data attributes. Bit 12 of the PMUFLAG field to indicate this. See Table H.7 .
ANALOG	ANALOG	No change.
DIGITAL	DIGITAL	No change.

H.4.5 STAT word conversion from version 2 to version 3

A substation PDC may construct the outgoing version 3 STAT_FLAG word from the incoming version 2 STAT word as shown in [Table H.9](#).

Table H.9—Partial conversion of version 2 STAT word to version 3

Outgoing Version 3	Description	Action to take on incoming version 2 STAT word
Bits		
15 to 14	Data error	Copy from bits 15 to 14 of incoming STAT word.
13	PMU sync	Calculated based on incoming STAT word, bits (8 to 6) (PMU_TQ) 0 = PMU_TQ code is 0 or 1. 1 = PMU_TQ code is greater than 1.
12	Data sorting	Copy from bit 12 of incoming STAT word.
11	Data bad	Set this bit normally for version 3.
10	Inserted data	Set this bit normally for version 3.
9 to 0	Reserved	Set to 0.

Annex I

(informative)

CONT_IDX handling example

I.1 Description

Fragmentation can be performed by multiple methods. Three examples are as follows:

- All fragments (except the last) are sized at 65 535 B. The last fragment includes the remaining bytes of the total frame.
- All fragments may be sized differently, each smaller than or equal to 65 535, broken up at a convenient boundary, such as a complete PMU section, or a signal name.
- Each fragment may be smaller than the maximum transmission unit to improve speed and reliability of transmission.

It should be noted that any method other than the first will reduce the total possible size of the configuration frame to less than the maximum (4 GB). This should not be a problem in most cases.

The first method of fragmenting Configuration frames is described in this section. A virtual configuration frame of required size is created in memory in accordance with [Table 3](#), except the following:

- FRAMESIZE field is set to 65 535, instead of the actual, larger size.
- CONT_IDX field is set to “one.”
- CHK field represents CRC-CCITT for the entire virtual frame.

The contiguous virtual frame is then fragmented into consecutive 65K fragments for wire transmission. Each transmitted fragment has its own header so it can be identified for recombination by the receiver. Each transmitted frame includes the following:

- A 16-B header, consisting of all fields up to and including CONT_IDX field (fields 1 to 7 in [Table 3](#)).
- FRAMESIZE field is always set to 65 535, except for last frame which is smaller, depending on the size of the last fragment.
- Consecutive frames have CONT_IDX field incrementing by one, except for the last transmitted frame, where it is set to 65535 (0xFFFF).
- All header fields other than FRAMESIZE and CONT_IDX match the initial frame.
- First transmitted frame will be the first 65 535 B of the virtual frame.
- All subsequent transmitted frames, except the last one, consist of 16 B of header and the next 65 519 unaligned bytes of the virtual frame. Thus, the size of each is 65 535 (= 16 B of header + 65 519 B of unaligned data).
- The last transmitted frame includes the header, and remainder of unaligned data. This frame includes the CHK field. Its FRAMESIZE field will be (16 + size of remaining unaligned data). Its CONT_IDX is 65535 (0xFFFF).

Received fragmented frames are reassembled in memory back into a virtual frame. The first frame is included in its entirety. Subsequent frames are added to the virtual frame from the first byte beyond the 16-B of the

header. Only the last frame will include the CHK field value that represents the CRC for the entire virtual frame. Having a CHK field only in the last frame helps prevent requiring the field in every fragmented frame, thereby reducing complexity, and saving space. CRC validation of the entire virtual frame only happens once, after final recombination.

I.2 Calculating total spanned frames

Implementations of fragmented configuration frames will need to calculate the total number of required spanned frames. The following provides guidance on this calculation. These calculations will change if the fragment size is smaller than 65 535, or for unevenly sized fragments.

A contiguous virtual frame larger than 65K will require spanning:

```
let virtualFrameLength = total virtual frame bytes
```

Each spanned frame requires its own 16-B header including the CONT_IDX field. This means space for available data is 65 535 total bytes available minus 16 B of header:

```
let maxFrameDataLength = 65519 B
```

Virtual frame has its own initial header, so for the first frame, header is included. Total remaining bytes beyond first frame are then virtual frame length minus the first 65 535 B:

```
let lengthAfterFirst = virtualFrameLength - 65535 B
```

The last frame size may be less than 65K, so calculate the remainder amount, if any:

```
let lastFrameLength = lengthAfterFirst % maxFrameDataLength
```

Total spanned frames will then be the one initial frame *plus* remaining bytes beyond first frame divided by maximum frame data length (using integer division), *plus* one final frame, if there is any remainder:

```
let totalFrames = 1 + lengthAfterFirst / maxFrameDataLength
```

```
if lastFrameLength > 0 then totalFrames = totalFrames + 1
```

Annex J

(informative)

XML format config frames

Configuration frames defined in 4.4 consist of binary data which are not human readable. This annex defines an XML format for configuration details in a human readable format for metadata exchange between users and archival. It is not intended as a substitute for binary configuration frames between devices.

```
<?xml version="1.0" encoding="utf-8"?>
<STREAM>
  <STREAMID>value</STREAM_ID>
  <TIME_BASE>value</TIME_BASE>
  <PDC_NAME>value</PDC_NAME>
  <NUM_PMU>value</NUM_PMU>
  <PMUS>
    <PMU1>
      <PMU_NAME>value</PMU_NAME>
      <PMU_ID>value</PMU_ID>
      <PMU_VERSION>value</PMU_VERSION>
      <G_PMU_ID>value</G_PMU_ID>
      <FORMAT>0Xvalue</FORMAT>
      <PHNMR>value</PHNMR>
      <ANNMR>value</ANNMR>
      <FRNMR>value</FRNMR>
      <DFDTNMR>value</DFDTNMR>
      <DGNMR>value</DGNMR>
      <PHASORS>
        <PH1>
          <NAME>value</NAME>
          <FLAG>0Xvalue</FLAG>
          <TYPE>0Xvalue</TYPE>
          <USER>0Xvalue</USER>
          <SCALING>value</SCALING>
          <ANGLE_OFFSET>value</ANGLE_OFFSET>
          <VOLTAGE_CLASS>value</VOLTAGE_CLASS>
        </PH1>
        .
      </PHASORS>
      <FREQ>
        <FR1>
          <NAME>value</NAME>
          <SCALING>value</SCALING>
          <OFFSET>value</OFFSET>
        </FR1>
        .
      </FREQ>
      <DFDT>
        <DFDT1>
          <NAME>value</NAME>
```

```

        <OFFSET>value</OFFSET>
    </DFDT1>
    .
    .
</DFDT>
<ANALOG>
    <AN1>
        <NAME>value</NAME>
        <SCALING>value</SCALING>
        <OFFSET>value</OFFSET>
    </AN1>
    .
    .

</ANALOG>
<DIGITAL>
    <DIG1>
        <BIT0><NAME>value</NAME></BIT0>
        .
        .
        <BIT15><NAME>value</NAME></BIT15>
    <DIGUNIT>
        <STATUS>0Xvalue</STATUS>
        <INPUT>0Xvalue</INPUT>
    </DIGUNIT>
</DIG1>
    .
    .
</DIGITAL>
<PMU_LAT>value</PMU_LAT>
<PMU_LON>value</PMU_LON>
<PMU_ELEV>value</PMU_ELEV>
<PMUFLAG>0Xvalue</PMUFLAG>
<WINDOW>0Xvalue</WINDOW>
<GRP_DLY>0Xvalue</GRP_DLY>
<PMU_DATA_RATE>0Xvalue</PMU_DATA_RATE>
<CFGCNT>0Xvalue</CFGCNT>
<USER_DEFINED>
    .
</USER_DEFINED>
</PMU1>
    .
    .
</PMUS>
    .
    .
    <STREAM_DATA_RATE>value</STREAM_DATA_RATE>
    <WAIT_TIME>value</WAIT_TIME>
</STREAM>
<USER_DEFINED>
    .
</USER_DEFINED>

```

Annex K

(informative)

Bibliography

Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

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